Splitting Idempotents via Interpolation in Multiplicative Linear Logic

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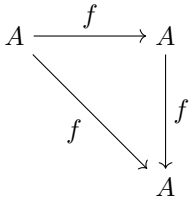
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Abstract. This paper shows that every idempotent splits in multiplicative linear logic (with and without the *mix*-rules): every idempotent can be written as a section composed with a retraction. Categorically speaking, the free $\star$-autonomous category is idempotent complete. Interestingly, this decomposition can be directly obtained by applying on the given idempotent an algorithm for proof-relevant interpolation.

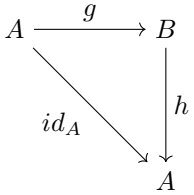
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1 Introduction

Idempotents are functions that "square to themselves". They can be broadly defined in category theory: a morphism $A \xrightarrow{f} A$ is an idempotent when $f \circ f = f$, *i.e.* when the following diagram commutes:



Idempotents are studied as a way to better understand the category under consideration. It is well-known that an idempotent can be built from any section-retraction pair. A section-retraction pair is the data of two morphisms $A \xrightarrow{g} B$ and $B \xrightarrow{h} A$ such that $h \circ g = id_A$, *i.e.* such that the following commutes:



Given a section-retraction pair (g, h), the morphism $g \circ h$ is an idempotent:

$$(g \circ h) \circ (g \circ h) = g \circ ((h \circ g) \circ h) = g \circ (id_A \circ h) = g \circ h$$

An idempotent $g \circ h$ built this way is called a split idempotent.

In some categories, every idempotent is a split idempotent: this property of the category is called being **idempotent complete** (*a.k.a.* Karoubi complete, Cauchy complete). Categories that are idempotent complete have some good properties [4]: the splitting of an idempotent is both a limit and a colimit, is absolute – *i.e.* is preserved by any functor – and every absolute (co)limit can be constructed from split idempotents. Examples of idempotent complete frameworks include the category Set (folklore), some homotopy (sub-)categories [22], and some triangulated categories [3]. However, not every category is idempotent complete, *e.g.* Martin-Löf type theory has idempotents that do not split [24]. Let us also point out that there exists a completion procedure yielding an idempotent complete category starting from any small category [4, Theorem 1].

We consider idempotents and split idempotents in the multiplicative fragment of linear logic [13]. This is one of the, if not the, main sub-systems of linear logic that are very well-behaved but still quite expressive. In order to get a category associated to this logic, following what happens in λ-calculus through the Curry-Howard-Lambek correspondence, its proofs are considered up to *cut-elimination* and *axiom-expansion* – which are the logical counterparts of β-reduction and η-expansion. This is the usual quotient, denoted by $=_{\beta\eta}$, of multiplicative linear logic to obtain the free $\star$-autonomous category whose objects are formulas and whose morphisms are equivalence classes of proofs – see *e.g.* [23]. In that setting, composition of a proof π of $\vdash A^\perp, B$ and a proof ϕ of $\vdash B^\perp, C$ is done through the *cut*-rule, yielding $\dfrac{\overset{\pi}{\vdash A^\perp, B} \quad \overset{\phi}{\vdash B^\perp, C}}{\vdash A^\perp, C}\ (cut)$; the identity morphism is (the equivalence class of) the proof made of a single *ax*-rule $\dfrac{}{\vdash A^\perp, A}\ (ax)$. This categorical model permits to instantiate the definitions of an idempotent and of a section-retraction pair to the sequent calculus of multiplicative linear logic.

Definition 1 ((Split) Idempotent, Section-Retraction pair).

– *An* ***idempotent*** *is a proof τ of $\vdash A^\perp, A$ such that:*

$$\frac{\overset{\tau}{\vdash A^\perp, A} \quad \overset{\tau}{\vdash A^\perp, A}}{\vdash A^\perp, A}\ (cut) \quad =_{\beta\eta} \quad \tau \tag{1}$$

– *A* ***section-retraction pair*** *is a pair of proofs π and ϕ, respectively of $\vdash A^\perp, B$ and of $\vdash B^\perp, A$, such that:*

$$\frac{\overset{\phi}{\vdash B^\perp, A} \quad \overset{\pi}{\vdash A^\perp, B}}{\vdash B^\perp, B}\ (cut) \quad =_{\beta\eta} \quad \frac{}{\vdash B^\perp, B}\ (ax) \tag{2}$$

– *A* ***split idempotent*** *is a proof* τ *such that* $\tau =_{\beta\eta} \dfrac{\overset{\pi}{\vdash A^\perp, B} \qquad \overset{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A}\ (cut)$
for some section-retraction pair (π, ϕ).

We prove in this paper that, in multiplicative linear logic, every idempotent is a split idempotent, so that idempotents are exactly split idempotents. Taking a categorical viewpoint, we establish that the free $\star$-autonomous category is idempotent complete. How to obtain the section-retraction pair (π, ϕ) from τ is surprisingly straightforward: it almost suffices to apply to τ the algorithm from Alexis Saurin's proof-relevant interpolation [21, Theorem 5]. More exactly, we use a variant of [21, Theorem 5] yielding hypotheses adequate to our study of idempotents, but whose proof is quite similar.

Theorem 2. *Let* τ *be a proof of a sequent* $\vdash A^\perp, B$ *in second-order linear logic (or in any of its fragments). There exists (in this fragment) a formula* C *and proofs* π *and* ϕ *of respectively* $\vdash A^\perp, C$ *and* $\vdash C^\perp, B$ *such that:*

(1) $\dfrac{\overset{\pi}{\vdash A^\perp, C} \qquad \overset{\phi}{\vdash C^\perp, B}}{\vdash A^\perp, B}\ (cut)$ *reduces by cut-elimination to* τ*;*
(2) *if* τ *is cut-free, then so are* π *and* ϕ*;*
(3) *if* τ *is ax-expanded, then so are* π *and* ϕ*;*
(4) *each ax-rule in* π *(respectively* ϕ*) uses at least one sub-formula occurrence from* $A^\perp$ *(respectively from* B*);*
(5) *each* 1*-rule in* π *(respectively* ϕ*) whose main formula is a* 1 *from* C *(respectively from* $C^\perp$*) is directly above a* $\perp$*-rule whose main formula is a* $\perp$ *from* $A^\perp$ *(respectively from* B*);*
(6) *each* 0 *from* C *(respectively from* $C^\perp$*) is in* π *(respectively in* ϕ*) in the context of a* $\top$*-rule whose main formula is a* $\top$ *from* $A^\perp$ *(respectively from* B*).*

Proof. Similar to [21]. See Appendix B for a detailed proof. □

Thanks to Theorem 2, to prove an idempotent τ is split it suffices to show that the two resulting interpolation proofs π and ϕ form a section-retraction pair. This is what we establish in this paper. Notice that the *proof-relevant* character, *i.e.* Item (1), is crucial here. The usual Craig's and Lyndon's interpolations are useless for our purposes, since an idempotent τ is a proof of $\vdash A^\perp, A$ so that looking only at formulas cannot yield anything interesting: if we pay no heed to the proof τ itself, then it could as well be the identity and it is hopeless to decompose τ simply by interpolating the identity. Also, please observe that such an interpolation result is, in the propositional case, stronger than Lyndon's interpolation, hence than Craig's interpolation too, simply using Item (4). The concept of proof-relevant interpolation is quite recent, and this paper gives a nice and maybe unexpected application of this new notion: splitting idempotents. The strengthening of Theorem 2 compared to [21, Theorem 5] consists of Items (3) to (6). Let us point out that Items (3), (4) and (6) were already given by the

algorithm proving [21, Theorem 5], even if not stated per se; however, Item (5) was not. Besides, the restriction on *ax*-rules in Item (4) was already considered when studying section-retraction pairs [5].

Since we work in multiplicative linear logic, it would make sense to use not the sequent calculus syntax but the proof-net syntax [15], which is a graphical syntax for proofs of this logic. Indeed, this syntax is more canonical than sequent calculus concerning unit-free multiplicative linear logic: two proofs are equal up to $=_{\beta\eta}$, that is two proofs are the same, exactly when they have the same normal proof-net (this follows from [18, Theorem 1] and [5, Theorem 2.50]). Furthermore, Alexis Saurin extended his proof-relevant interpolation theorem to proof-nets working with Guido Fiorillo and Daniel Osorio Valencia [10].[1] Nonetheless, we chose not to do so for two reasons. First, while it is possible to consider units in proof-nets, they are not so easy to manipulate and typically involve an equivalence relation such as Trimble rewiring [2]. Second, we hope to extend our result beyond the framework of multiplicative linear logic by adding additive connectives and units, or exponential connectives, or quantifiers, in which cases proof-nets are not as canonical; having the easier multiplicative case in sequent calculus should make such extensions more direct. Still, since intuitions might be clearer using proof-nets, the key points in our argumentation will be illustrated by graphs; and for the reader familiar with proof-nets it should be most easy to translate our demonstration from sequent calculus to the proof-net syntax.

Outline. We begin by quickly recalling the sequent calculus syntax of multiplicative linear logic in Section 2. Afterwards, we introduce in Section 3 some useful but technical results used to decide when two proofs are the same up to cut-elimination and axiom-expansion. These results are applied in the main part, Section 4, to prove that the proofs resulting from the interpolation of an idempotent form a section-retraction pair. We end with Section 5, a succinct part extending our main result to an extended framework with the *mix*-rules.

2 Multiplicative Linear Logic

We present here multiplicative linear logic in a standard way as a *unilateral* sequent calculus [13]. **Formulas** are defined by the following grammar, with X an atom in some countable set:

$$A, B ::= X \mid X^{\perp} \mid A \parr B \mid A \otimes B \mid \bot \mid 1$$

Formulas of the kind X or $X^{\perp}$ are called **atomic**. **Orthogonality** $(\cdot)^{\perp}$ (*a.k.a.* negation, duality) is extended as an involution to arbitrary formulas using the De Morgan's laws:

$$(X^{\perp})^{\perp} \stackrel{\text{def}}{=} X \quad (A \parr B)^{\perp} \stackrel{\text{def}}{=} A^{\perp} \otimes B^{\perp} \quad (A \otimes B)^{\perp} \stackrel{\text{def}}{=} A^{\perp} \parr B^{\perp} \quad \bot^{\perp} \stackrel{\text{def}}{=} 1 \quad 1^{\perp} \stackrel{\text{def}}{=} \bot$$

[1] That our version of interpolation, Theorem 2, also holds in the proof-net syntax follows by applying the algorithm expanding *cut* in [7, Appendix E] or by following the interpolation algorithm in [11].

$$\frac{}{\vdash A^\perp, A}\ (ax) \qquad \frac{\vdash A, \Gamma \quad \vdash A^\perp, \Delta}{\vdash \Gamma, \Delta}\ (cut)$$

$$\frac{\vdash A, B, \Gamma}{\vdash A \parr B, \Gamma}\ (\parr) \qquad \frac{\vdash A, \Gamma \quad \vdash B, \Delta}{\vdash A \otimes B, \Gamma, \Delta}\ (\otimes) \qquad \frac{\vdash \Gamma}{\vdash \bot, \Gamma}\ (\bot) \qquad \frac{}{\vdash 1}\ (1)$$

Fig. 1. Rules of Multiplicative Linear Logic

Rules of the logic are given in Figure 1, where A and B stand for arbitrary formulas, and Γ and Δ for sets of (occurrences of) formulas. This sequent calculus is equipped with a **cut-elimination** and an **axiom-expansion** procedures (not recalled here) that are weakly normalising [20], and whose normal forms are respectively *cut-free* proofs (with no *cut*-rule) and *ax-expanded* proofs (with *ax*-rules only on atomic formulas). Plus, cut-elimination preserves being an *ax*-expanded proof. **Equality up to cut-elimination and axiom-expansion** is denoted by $=_{\beta\eta}$.

We recall that **idempotents**, **split idempotents** and **section-retraction pairs** are defined in the introduction (*c.f.* Definition 1).

3 Occurrences of Atoms and Units in Normal Forms

We do not define properly occurrences, but use the following convention. We employ indexes to distinguish different occurrences of a same atom, and when X_i is an occurrence of X in some formula C, then $X_i^\perp$ is the *dual* occurrence of X_i in $C^\perp$ as identified by the De Morgan's laws defining $(\cdot)^\perp$. We proceed similarly with units.

To prove our main theorem, we will use some results about *ax*-rules, 1-rules, $\bot$-rules and occurrences that are either easy or that follow from the literature. Since those are not the main focus of this paper, their proofs are given in Appendix A. Besides, it will be easier to work up to isomorphism, in particular up to neutrality of the units.

Definition 3. *A formula is* ***distributed*** *if none of its sub-formulas are of one of the following forms (where A is any formula):*

$$A \otimes 1 \qquad 1 \otimes A \qquad A \parr \bot \qquad \bot \parr A$$

Remark that a formula A is distributed if and only if $A^\perp$ is distributed. Up to isomorphism, 1 is the unit of $\otimes$ and $\bot$ the unit of $\parr$, hence a formula can always be assumed to be distributed – see [1,8] for details. In the proofs we consider, 1- and $\bot$-rules will always belong to a specific pattern.

Definition 4. *We call a* $1/\bot$***-pattern*** *the following sub-proof:*

$$\frac{\dfrac{}{\vdash 1}\ (1)}{\vdash \bot, 1}\ (\bot)$$

Fact 5. *Let π be an* ax-expanded *proof of a sequent $\vdash \Gamma$. Then:*

- *each occurrence of an atomic formula from $\vdash \Gamma$ is in the conclusion of a unique ax-rule of π;*
- *each occurrence of 1 (respectively $\bot$) from $\vdash \Gamma$ is the main formula of a unique 1-rule (respectively $\bot$-rule) of π.*

Lemma 6. *Let A and B be* distributed *formulas such that $\vdash A^\bot, B$ and $\vdash B^\bot, A$ are both provable. If π is a* cut-free *and* ax-expanded *proof of $\vdash A^\bot, B$, then each 1-rule and each $\bot$-rule in π belongs to a $1/\bot$-pattern.*

Remark 7. Lemma 6 is the analogue in the sequent calculus of a result in proof-nets by Balat and Di Cosmo [1, Theorem 19].

It is easy to know whether two proofs of $\vdash A^\bot, A$ are equal.

Theorem 8. *Let π and ϕ be* cut-free and ax-expanded *proofs of a same sequent $\vdash A^\bot, A$ for A a* distributed *formula. Then $\pi =_{\beta\eta} \phi$ if and only if:*

- *the ax-rules of π and ϕ regroup the exact same occurrences of atoms; and*
- *the $1/\bot$-patterns of π and ϕ regroup the exact same occurrences of 1 and $\bot$.*

Example 9. Let us consider the three proofs π, ϕ and τ in Figure 2 of the sequent $\vdash A^\bot, A$ with the distributed $A \stackrel{\text{def}}{=} (1_1 \parr X_2) \parr (X_3 \parr 1_4)$ – remember indices are used to follow occurrences. The first proof π is the (full) axiom-expansion of the ax-rule on $\vdash A^\bot, A$. Through Theorem 8, we obtain that $\pi =_{\beta\eta} \phi \neq_{\beta\eta} \tau$.

Remark 10. The general problem of deciding whether two proofs are the same in multiplicative linear logic is PSPACE-complete [16]; typically, it involves deciding an equivalence such as Trimble rewiring [2]. What we show with Theorem 8 is that, when restricted to sequents $\vdash A^\bot, A$ (up to isomorphism), this problem has the same complexity as the unit-free case, that can be solved in linear time. This generalises to distributed sequents $\vdash \Gamma$ containing as many (or more) 1 sub-formulas as (than) $\bot$ sub-formulas. In particular, this property underlies the considerations on units in [1], that studies isomorphisms in multiplicative linear logic by reducing them to the unit-free case.

Lastly, we use the following to find rules in a normal form.

Fact 11. *Let π be a proof of a sequent $\vdash \Gamma$.*

- *Consider in π ax-rules $\dfrac{}{\vdash X_1^\bot, X_2}\,(ax)$, $\dfrac{}{\vdash X_2^\bot, X_3}\,(ax)$, $\ldots$, $\dfrac{}{\vdash X_{n-1}^\bot, X_n}\,(ax)$ where $X_1^\bot$ and X_n occur in Γ whereas $X_i^\bot$ and X_i are dual occurrences of a formula introduced by a cut-rule for all $1 < i < n$. There is an ax-rule $\dfrac{}{\vdash X_1^\bot, X_n}\,(ax)$ in all cut-free ϕ such that π reduces by cut-elimination to ϕ.*

$$\pi \overset{\text{def}}{=} \dfrac{\dfrac{\dfrac{\dfrac{}{\vdash 1_1}(1)}{\vdash \bot_1, 1_1}(\bot) \quad \dfrac{}{\vdash X_2^\bot, X_2}(ax)}{\vdash \bot_1 \otimes X_2^\bot, 1_1, X_2}(\otimes)}{\vdash \bot_1 \otimes X_2^\bot, 1_1 \parr X_2}(\parr) \quad \dfrac{\dfrac{\dfrac{}{\vdash X_3^\bot, X_3}(ax) \quad \dfrac{}{\vdash 1_4}(1)}{\vdash X_3^\bot \otimes \bot_4, X_3, 1_4}(\otimes)}{\vdash X_3^\bot \otimes \bot_4, X_3 \parr 1_4}(\parr)}{\dfrac{\vdash (\bot_1 \otimes X_2^\bot) \otimes (X_3^\bot \otimes \bot_4), 1_1 \parr X_2, X_3 \parr 1_4}{\vdash (\bot_1 \otimes X_2^\bot) \otimes (X_3^\bot \otimes \bot_4), (1_1 \parr X_2) \parr (X_3 \parr 1_4)}(\parr)}(\otimes)$$

$$\phi \overset{\text{def}}{=} \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\vdash 1_1}(1)}{\vdash \bot_1, 1_1}(\bot) \quad \dfrac{}{\vdash X_2^\bot, X_2}(ax)}{\vdash \bot_1 \otimes X_2^\bot, 1_1, X_2}(\otimes) \quad \dfrac{\dfrac{}{\vdash X_3^\bot, X_3}(ax) \quad \dfrac{}{\vdash 1_4}(1)}{\vdash X_3^\bot \otimes \bot_4, X_3, 1_4}(\otimes)}{\vdash (\bot_1 \otimes X_2^\bot) \otimes (X_3^\bot \otimes \bot_4), 1_1, X_2, X_3, 1_4}(\otimes)}{\vdash (\bot_1 \otimes X_2^\bot) \otimes (X_3^\bot \otimes \bot_4), 1_1, X_2, X_3 \parr 1_4}(\parr)}{\vdash (\bot_1 \otimes X_2^\bot) \otimes (X_3^\bot \otimes \bot_4), 1_1 \parr X_2, X_3 \parr 1_4}(\parr)}{\vdash (\bot_1 \otimes X_2^\bot) \otimes (X_3^\bot \otimes \bot_4), (1_1 \parr X_2) \parr (X_3 \parr 1_4)}(\parr)$$

$$\tau \overset{\text{def}}{=} \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\vdash 1_4}(1)}{\vdash \bot_1, 1_4}(\bot) \quad \dfrac{}{\vdash X_2^\bot, X_2}(ax)}{\vdash \bot_1 \otimes X_2^\bot, 1_4, X_2}(\otimes) \quad \dfrac{\dfrac{}{\vdash X_3^\bot, X_3}(ax) \quad \dfrac{}{\vdash 1_1}(1)}{\vdash X_3^\bot \otimes \bot_4, X_3, 1_1}(\otimes)}{\vdash (\bot_1 \otimes X_2^\bot) \otimes (X_3^\bot \otimes \bot_4), 1_1, X_2, X_3, 1_4}(\otimes)}{\vdash (\bot_1 \otimes X_2^\bot) \otimes (X_3^\bot \otimes \bot_4), 1_1, X_2, X_3 \parr 1_4}(\parr)}{\vdash (\bot_1 \otimes X_2^\bot) \otimes (X_3^\bot \otimes \bot_4), 1_1 \parr X_2, X_3 \parr 1_4}(\parr)}{\vdash (\bot_1 \otimes X_2^\bot) \otimes (X_3^\bot \otimes \bot_4), (1_1 \parr X_2) \parr (X_3 \parr 1_4)}(\parr)$$

Fig. 2. Proofs of a same sequent, with $\pi =_{\beta\eta} \phi \neq_{\beta\eta} \tau$

– *Consider in* π $1/\bot$*-patterns* $\dfrac{\dfrac{}{\vdash 1_2}\,{}^{(1)}}{\vdash \bot_1, 1_2}\,{}^{(\bot)}$, $\dfrac{\dfrac{}{\vdash 1_3}\,{}^{(1)}}{\vdash \bot_2, 1_3}\,{}^{(\bot)}$, $\ldots$, $\dfrac{\dfrac{}{\vdash 1_n}\,{}^{(1)}}{\vdash \bot_{n-1}, 1_n}\,{}^{(\bot)}$ *where* $\bot_1$ *and* 1_n *occur in* Γ *whereas* $\bot_i$ *and* 1_i *are dual occurrences of a formula introduced by a cut-rule in* π *for all* $1 < i < n$*. There is a* $1/\bot$*-pattern* $\dfrac{\dfrac{}{\vdash 1_n}\,{}^{(1)}}{\vdash \bot_1, 1_n}\,{}^{(\bot)}$ *in some cut-free* ϕ *such that* π *reduces by cut-elimination to* ϕ*.*

4 Splitting an Idempotent

We prove here our main result: applying the proof-relevant interpolation theorem on an idempotent yields a section-retraction pair, henceforth every idempotent is split. In all this section, we assume given an idempotent τ of the sequent $\vdash A^\bot, A$. Without any loss of generality, we assume that τ is *cut*-free and *ax*-expanded, and that A is distributed. By applying Theorem 2 on τ, we obtain a formula B (distributed *w.l.o.g.*) and two *cut*-free and *ax*-expanded proofs π and ϕ, respectively of $\vdash A^\bot, B$ and of $\vdash B^\bot, A$, such that:

(a) $\dfrac{\overset{\pi}{\vdash A^\bot, B} \qquad \overset{\phi}{\vdash B^\bot, A}}{\vdash A^\bot, A}\,{}^{(cut)}$ reduces by cut-elimination to τ;
(b) no *ax*-rule in π (respectively ϕ) uses two atomic formulas occurring in B (respectively in $B^\bot$);
(c) no $1/\bot$-pattern in π (respectively ϕ) uses two units occurring in B (respectively in $B^\bot$).

Our end goal is proving that these proofs π and ϕ form a section-retraction pair. More precisely, we show that $\dfrac{\overset{\phi}{\vdash B^\bot, A} \qquad \overset{\pi}{\vdash A^\bot, B}}{\vdash B^\bot, B}\,{}^{(cut)}$ is equal, up to cut-elimination, to the full axiom-expansion of $\dfrac{}{\vdash B^\bot, B}\,{}^{(ax)}$.[2] It is hence natural to consider an arbitrary normal – *i.e. cut*-free – form μ of $\dfrac{\overset{\phi}{\vdash B^\bot, A} \qquad \overset{\pi}{\vdash A^\bot, B}}{\vdash B^\bot, B}\,{}^{(cut)}$ and to prove through Theorem 8 that μ is an axiom-expansion of $\dfrac{}{\vdash B^\bot, B}\,{}^{(ax)}$ (or a proof equal to it). Henceforth, we establish the following results about μ.

Lemma 12.

– *In each ax-rule* $\dfrac{}{\vdash X^\bot, X}\,{}^{(ax)}$ *of* μ *(for any atom* X*), one of* $X^\bot$ *and* X *occurs in* $B^\bot$ *and the other in* B*.*
– *In each* $1/\bot$*-pattern of* μ*, one of* 1 *and* $\bot$ *occurs in* $B^\bot$ *and the other in* B*.*

[2] As an example of simplification when using proof-nets, in that setting one can replace "is equal up to cut-elimination" with "reduces by cut-elimination".

Lemma 13.

- *Consider an ax-rule* $\dfrac{}{\vdash X^\perp, X}\,{}^{(ax)}$ *of* μ *where either* $X^\perp$ *or* X *occurs in* $B^\perp$ *and the other in* B*. Then* $X^\perp$ *and* X *are dual occurrences in* $B^\perp$ *and* B*.*
- *Consider a* $1/\perp$*-pattern of* μ *where either* 1 *or* $\perp$ *occurs in* $B^\perp$ *and the other in* B*. Then* 1 *and* $\perp$ *are dual occurrences in* $B^\perp$ *and* B*.*

We proceed to prove these intermediate results, then conclude with the demonstration of our main theorem. We will use Items (b) and (c) in our demonstrations, as well as the following:

(d) the normal forms of $\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B} \quad \overset{\mu}{\vdash B^\perp, B}}{\vdash A^\perp, B}\,{}^{(cut)} \qquad \overset{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A}\,{}^{(cut)}$ and those of $\dfrac{\overset{\pi}{\vdash A^\perp, B} \quad \overset{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A}\,{}^{(cut)}$ are all equal to τ up to $=_{\beta\eta}$.

Item (d) is our sole use of the idempotency of τ and of Item (a). Its proof is a simple computation:

$$\dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B} \quad \overset{\mu}{\vdash B^\perp, B}}{\vdash A^\perp, B}\,{}^{(cut)} \qquad \overset{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A}\,{}^{(cut)}$$

$$=_{\beta\eta} \quad \dfrac{\overset{\pi}{\vdash A^\perp, B} \qquad \dfrac{\overset{\phi}{\vdash B^\perp, A} \quad \overset{\pi}{\vdash A^\perp, B}}{\vdash B^\perp, B}\,{}^{(cut)}}{\vdash A^\perp, B} \qquad \overset{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A}\,{}^{(cut)}$$

by definition of μ

$$=_{\beta\eta} \quad \dfrac{\dfrac{\overset{\pi}{\vdash A^\perp, B} \quad \overset{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A}\,{}^{(cut)} \qquad \dfrac{\overset{\pi}{\vdash A^\perp, B} \quad \overset{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A}\,{}^{(cut)}}{\vdash A^\perp, A}\,{}^{(cut)}$$

by associativity of *cut*

$$=_{\beta\eta} \quad \dfrac{\overset{\pi}{\vdash A^\perp, B} \quad \overset{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A}\,{}^{(cut)} \quad =_{\beta\eta} \tau$$

by idempotence of τ and Item (a)

Let us start with the proof of Lemma 12.

Proof of Lemma 12. We present here only a proof of the first item. The second item can be proved the same way, simply by replacing positive atoms by 1, negative atoms by $\bot$, *ax*-rules by $1/\bot$ patterns, accompanying uses of Fact 5 by a call to Lemma 6, and replacing uses of Item (b) by uses of Item (c).

We proceed by *reductio ad absurdum*. Consider an *ax*-rule $\dfrac{}{\vdash X_1^\bot, X_2}\,{}^{(ax)}$ in μ with X_1 and $X_2^\bot$ both occurring in B – the case where they occur in $B^\bot$ is symmetric. We use Item (d) and Theorem 8 to find in τ incompatible *ax*-rules. See Figure 3 for a "proof-net illustration", where triangles represent formula trees and dashed or dotted edges stand for *ax*-rules between two occurrences.

We assumed in μ an *ax*-rule $\dfrac{}{\vdash X_1^\bot, X_2}\,{}^{(ax)}$ with X_1 and $X_2^\bot$ both occurring in B. Consider the occurrences X_1 and $X_2^\bot$ of $B^\bot$: by Fact 5, there are in ϕ *ax*-rules $\dfrac{}{\vdash X_1, X_3^\bot}\,{}^{(ax)}$ and $\dfrac{}{\vdash X_2^\bot, X_4}\,{}^{(ax)}$, with $X_3^\bot$ and X_4 both occurring in A by Item (b) – see the upper half of Figure 3. Therefore, in some normal form of $\dfrac{\dfrac{\overset{\pi}{\vdash A^\bot, B} \quad \overset{\mu}{\vdash B^\bot, B}}{\vdash A^\bot, B}\,{}^{(cut)} \quad \overset{\phi}{\vdash B^\bot, A}}{\vdash A^\bot, A}\,{}^{(cut)}$ there is an *ax*-rule $\dfrac{}{\vdash X_3^\bot, X_4}\,{}^{(ax)}$ by Fact 11, hence also in τ by Item (d) and Theorem 8.

Consider now $\dfrac{\overset{\pi}{\vdash A^\bot, B} \quad \overset{\phi}{\vdash B^\bot, A}}{\vdash A^\bot, A}\,{}^{(cut)}$. Recall we found in ϕ the *ax*-rules $\dfrac{}{\vdash X_1, X_3^\bot}\,{}^{(ax)}$ and $\dfrac{}{\vdash X_2^\bot, X_4}\,{}^{(ax)}$. Looking at the occurrences $X_1^\bot$ and X_2 of B, by Fact 5 there are in π *ax*-rules $\dfrac{}{\vdash X_1^\bot, X_5}\,{}^{(ax)}$ and $\dfrac{}{\vdash X_2, X_6^\bot}\,{}^{(ax)}$ with X_5 and $X_6^\bot$ both occurring in $A^\bot$ by Item (b) – see the lower half of Figure 3. Therefore, in some normal form of $\dfrac{\overset{\pi}{\vdash A^\bot, B} \quad \overset{\phi}{\vdash B^\bot, A}}{\vdash A^\bot, A}\,{}^{(cut)}$ there are *ax*-rules $\dfrac{}{\vdash X_3^\bot, X_5}\,{}^{(ax)}$ and $\dfrac{}{\vdash X_4, X_6^\bot}\,{}^{(ax)}$ by Fact 11, hence also in τ by Item (d) and Theorem 8.

We have reached a contradiction. In τ there is both an *ax*-rule $\dfrac{}{\vdash X_3^\bot, X_4}\,{}^{(ax)}$ and an *ax*-rule $\dfrac{}{\vdash X_3^\bot, X_5}\,{}^{(ax)}$, with X_4 occurring in A and X_5 in $A^\bot$ so that X_4 and X_5 are not the same occurrence. This is impossible through Fact 5. □

Proving Lemma 13 can be done through a similar reasoning.

Proof of Lemma 13. We present here only a proof of the first item. The second item can be proved the same way, simply by replacing positive atoms by 1, negative atoms by $\bot$, *ax*-rules by $1/\bot$ patterns, accompanying uses of Fact 5 by a call to Lemma 6, and replacing uses of Item (b) by uses of Item (c).

Consider in μ any *ax*-rule $\dfrac{}{\vdash X_1^\bot, X_2}\,{}^{(ax)}$ where $X_1^\bot$ occurs in $B^\bot$ and X_2 in B – the case where $X_1^\bot$ occurs in B and X_2 in $B^\bot$ is symmetric. Let us prove

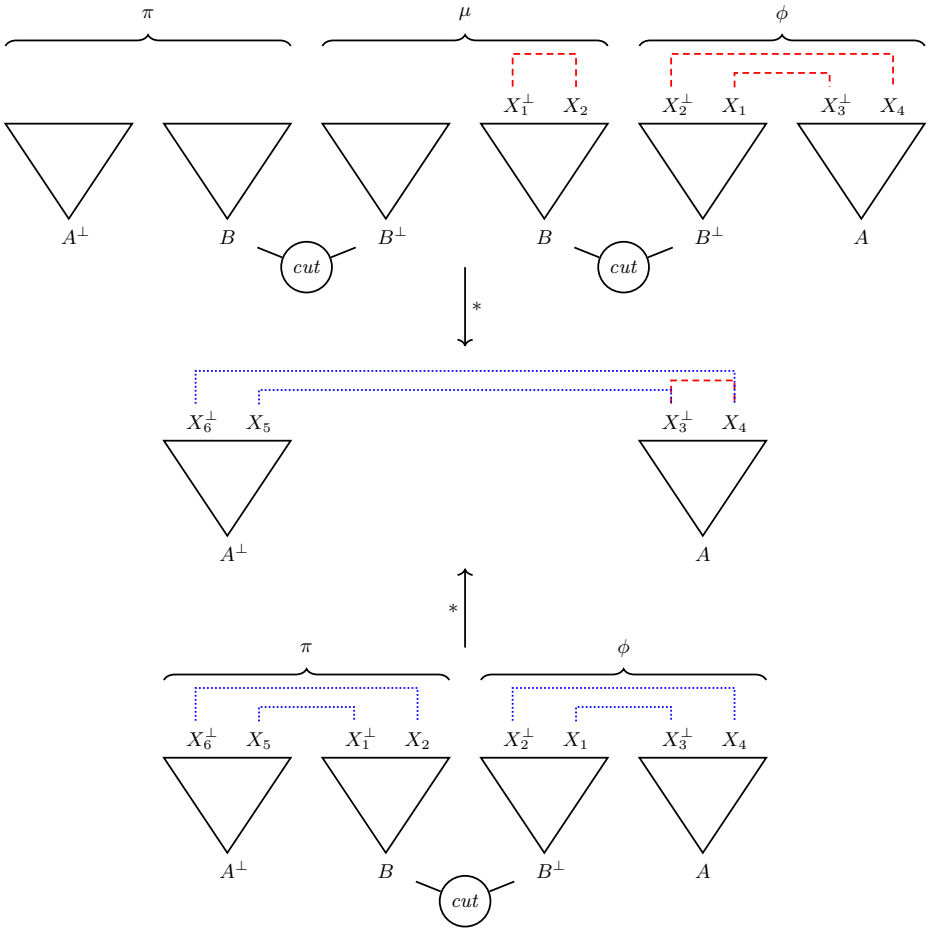


Fig. 3. Illustration of the *ax*-rules found in the proof of Lemma 12

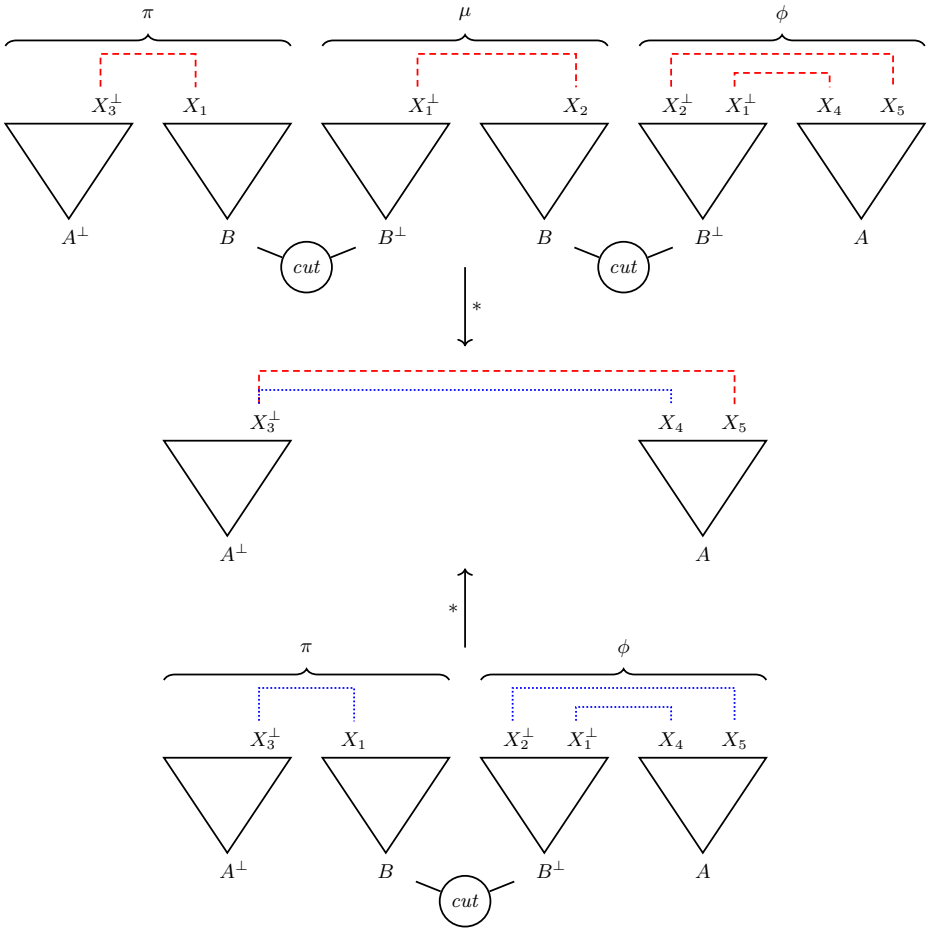


Fig. 4. Illustration of the ax-rules found in the proof of Lemma 13

that $X_1^\perp$ and $X_2^\perp$ refer to the same occurrence. Proceeding as in the proof of Lemma 12, we find three *ax*-rules using Fact 5 and Item (b) – see Figure 4 for a "proof-net illustration" (with the same conventions as for Figure 3):

- in π there is $\dfrac{}{\vdash X_3^\perp, X_1}\,{}^{(ax)}$ where $X_3^\perp$ occurs in $A^\perp$;
- in ϕ there is $\dfrac{}{\vdash X_1^\perp, X_4}\,{}^{(ax)}$ where X_4 occurs in A;
- in ϕ there is $\dfrac{}{\vdash X_2^\perp, X_5}\,{}^{(ax)}$ where X_5 occurs in A.

Consider first the proof $\dfrac{\overset{\pi}{\vdash A^\perp, B} \qquad \dfrac{\overset{\mu}{\vdash B^\perp, B} \qquad \overset{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, B}\,{}^{(cut)}}{\vdash A^\perp, A}\,{}^{(cut)}$. Following the chain of *ax*-rules $\dfrac{}{\vdash X_3^\perp, X_1}\,{}^{(ax)}$ (in π), $\dfrac{}{\vdash X_1^\perp, X_2}\,{}^{(ax)}$ (in μ), and $\dfrac{}{\vdash X_2^\perp, X_5}\,{}^{(ax)}$ (in ϕ), there is an *ax*-rule $\dfrac{}{\vdash X_3^\perp, X_5}\,{}^{(ax)}$ in τ thanks to Fact 11, Item (d), and Theorem 8 – see the upper half of Figure 4.

Now, let us look at $\dfrac{\overset{\pi}{\vdash A^\perp, B} \qquad \overset{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A}\,{}^{(cut)}$. Following the chain of *ax*-rules $\dfrac{}{\vdash X_3^\perp, X_1}\,{}^{(ax)}$ (in π), and $\dfrac{}{\vdash X_1^\perp, X_4}\,{}^{(ax)}$ (in ϕ), there is an *ax*-rule $\dfrac{}{\vdash X_3^\perp, X_4}\,{}^{(ax)}$ in τ thanks to Fact 11, Item (d), and Theorem 8 – see the lower half of Figure 4.

To sum up, we have $\dfrac{}{\vdash X_3^\perp, X_5}\,{}^{(ax)}$ and $\dfrac{}{\vdash X_3^\perp, X_4}\,{}^{(ax)}$ in τ. Since a unique *ax*-rule contains the occurrence $X_3^\perp$ by Fact 5, X_5 is the same occurrence as X_4. But then, using $\dfrac{}{\vdash X_1^\perp, X_4}\,{}^{(ax)}$ and $\dfrac{}{\vdash X_2^\perp, X_5}\,{}^{(ax)}$ in ϕ with the same argument, it follows that $X_1^\perp$ and $X_2^\perp$ are the same occurrence. This concludes the proof. □

Putting everything together, we obtain our main result.

Theorem 14 (MLL is idempotent complete). *The free $\star$-autonomous category is idempotent complete. In other words, in multiplicative linear logic given an idempotent τ of conclusion $\vdash A^\perp, A$ there exists a section-retraction pair (π, ϕ) such that $\tau =_{\beta\eta} \dfrac{\overset{\pi}{\vdash A^\perp, B} \qquad \overset{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A}\,{}^{(cut)}$.*

Proof. It is well known that multiplicative linear logic is the internal logic of the free $\star$-autonomous category – see *e.g.* [19, Theorem 2.3.1], [17, Theorem 3], [23, Proposition 1.5], and [2, Theorem 5.1] – so that proofs of multiplicative linear logic up to cut-elimination and axiom-expansion (or equivalently, up to rules commutation [5, Theorem 2.50]) are the same as morphisms of the free $\star$-autonomous category.

Without any loss of generality, we assume that τ is *cut*-free and *ax*-expanded. Were it not the case, we consider a normal form τ' of τ instead, using that

$\tau' =_{\beta\eta} \tau$. Theorem 2 applied on τ yields a formula B and two proofs π and ϕ respectively of $\vdash A^\perp, B$ and of $\vdash B^\perp, A$. We can assume without loss of generality that A and B are both distributed – see Appendix C for details. Therefore $\dfrac{\overset{\pi}{\vdash A^\perp, B} \quad \overset{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A}\,{}_{(cut)} =_{\beta\eta} \tau$ and, by Theorem 8 and Lemmas 12 and 13, $\dfrac{\overset{\phi}{\vdash B^\perp, A} \quad \overset{\pi}{\vdash A^\perp, B}}{\vdash B^\perp, B}\,{}_{(cut)} =_{\beta\eta} \dfrac{}{\vdash B^\perp, B}\,{}^{(ax)}$ (using that the ax-rules and $1/\perp$ patterns of the full axiom-expansion of $\dfrac{}{\vdash B^\perp, B}\,{}^{(ax)}$ are exactly between dual occurrences from $B^\perp$ and B, which is easily provable by induction over B). Thus (π, ϕ) is a section-retraction pair, and τ is a split idempotent. □

5 Adding the *mix*-rules

Multiplicative linear logic is often extended with (some of) the ***mix*-rules** [12]:

$$\frac{\vdash \Gamma \quad \vdash \Delta}{\vdash \Gamma, \Delta}\,{}_{(mix_2)} \qquad \frac{}{\vdash}\,{}^{(mix_0)}$$

In presence of both the mix_2- and mix_0-rules, it is usually assumed that the second rule is neutral for the first, which amounts to extending equality with:

$$\frac{\dfrac{}{\vdash}\,{}^{(mix_0)} \quad \overset{\pi}{\vdash \Gamma}}{\vdash \Gamma}\,{}_{(mix_2)} =_{\beta\eta} \overset{\pi}{\vdash \Gamma} =_{\beta\eta} \frac{\overset{\pi}{\vdash \Gamma} \quad \dfrac{}{\vdash}\,{}^{(mix_0)}}{\vdash \Gamma}\,{}_{(mix_2)} \tag{3}$$

We call this identification the ***mix*-Retoré equality**, because *mix*-Retoré is the rewriting system towards the middle proof [9].

We establish quite easily that such extensions of multiplicative linear logic are idempotent complete, thanks to Theorem 14.

Theorem 15 (MLL with *mix* is idempotent complete). *Multiplicative linear logic extended with the mix_2-rule and/or the mix_0-rule and/or mix-Retoré equality is idempotent complete.*

Proof. The extensions to consider are the followings:

(a) the mix_2-rule;
(b) the mix_0-rule;
(c) the mix_2- and mix_0-rules;
(d) the mix_2- and mix_0-rules and *mix*-Retoré equality.

In each of these cases, the key point is that (classes of) idempotents are the same as in multiplicative linear logic without *mix*.

Case (a), Case (b) and Case (c) are almost immediate. Indeed, equality up to cut-elimination and axiom-expansion preserves the numbers of mix_2- and

mix_0-rules. Therefore, given an idempotent, Equation (1) implies its numbers of mix_2-rules and of mix_0-rules are equal to their double, so they can only be null. Henceforth, idempotents in these cases are exactly those of multiplicative linear logic with no *mix*-rule, which are split by Theorem 14.

(Notice this uses that, between *mix*-free proofs, equality up to cut-elimination and axiom-expansion is the same whether there are some *mix*-rules or not, which is a consequence of cut-elimination and axiom-expansion preserving the presence of *mix*-rules.)

Case (d) is subtler. Whereas *mix*-Retoré equality does not preserve the numbers of mix_2- and mix_0-rules, it preserves their difference. Henceforth, given a *cut*-free idempotent τ of $\vdash A^\perp, A$, it must have as many mix_2-rules as mix_0-rules by reasoning as in the preceding cases. We will prove that without loss of generality there is no mix_0-rule in τ, which will then be enough to conclude.

In that framework, it is known that the units are isomorphic: $1 \simeq \perp$ – see *e.g.* [5]. Hence, by working up to isomorphism, either A is unit-free or it is 1. The second case is trivial: the only (equivalence class of) proof μ of $\vdash \perp, 1$ is the identity $\dfrac{\dfrac{}{\vdash 1}\,{\scriptstyle (1)}}{\vdash \perp, 1}\,{\scriptstyle (\perp)}$ and it is a split idempotent: (μ, μ) is the corresponding section-retraction pair. Thus, let us assume that A is unit-free. In that case, the sole possible rule that can be directly below a mix_0-rule in τ is a mix_2-rule. We can assume it never happens by working up to *mix*-Retoré equality, so that τ contains no mix_0-rule – remark τ cannot be reduced to the mix_0-rule as its conclusion sequent is not empty. Thence, there is no mix_0-rule in τ, and as it contains as many mix_2-rules as mix_0-rules, there is no mix_2-rule either. This implies that τ is an idempotent in multiplicative linear logic with no *mix*-rule, hence is split by Theorem 14.

(Notice this uses that, between *mix*-free proofs of unit-free sequents, equality up to cut-elimination and axiom-expansion and *mix*-Retoré equality is simply equality up to cut-elimination and axiom-expansion of the *mix*-free setting; this follows from [5, Theorem 2.50] and the fact that no rule commutation in the unit-free case may put a rule below a mix_0-rule that is not a mix_2-rule that was already below it.) □

6 Conclusion

Given a section-retraction pair (π, ϕ), the proof $\dfrac{\overset{\pi}{\vdash A^\perp, B} \qquad \overset{\phi}{\vdash B^\perp, A}}{\vdash A^\perp, A}\,{\scriptstyle (cut)}$ is always an idempotent. The reciprocal holds in the framework of multiplicative linear logic: all idempotents are of this shape, *a.k.a.* are split idempotents (Theorem 14). Categorically speaking, the free $\star$-autonomous category is idempotent complete. This means that from section-retraction pairs one can obtain all idempotents, and conversely. Unfortunately, there is no exact characterisation of section-retraction pairs in multiplicative linear logic to this day – there only

is a characterisation with proofs up to cut-elimination only (not up to axiom-expansion) [5]. That all idempotents are split holds not only in multiplicative linear logic, but also extends in presence of (some of) the *mix*-rules (Theorem 15).

It would be interesting to check whether bigger fragments of linear logic are also idempotent complete. Notably, one may add additive connectives or units, or exponential connectives, or quantifiers. We think that adding the additive connectives & and $\oplus$ is easy: it should suffices to work slice by slice [13,15], proceeding in each slice as in the multiplicative fragment. The only delicate point seems to be proving that slices of the expanded *ax*-rule on $\vdash B^{\perp}, B$ are in bijection with slices *that are preserved during cut-elimination* of the composition by *cut* of π and ϕ with as *cut*-formula B. First-order quantifiers also may be not too complex to add, using the well-behaved notion of proof-nets [14]. On the other hand, exponentials seem hard to add, except in presence of the *mix*-rules.

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A Proofs for Comparing Normal Forms

This section presents proofs of the results claimed in Section 3. As explained there, these results are easy or follow from the literature. We first consider those about comparing normal forms, and then those for finding rules in normal forms.

A.1 Comparing Normal Forms

Fact 5. *Let π be an ax-*expanded *proof of a sequent $\vdash \Gamma$. Then:*

- *each occurrence of an atomic formula from $\vdash \Gamma$ is in the conclusion of a unique ax-rule of π;*
- *each occurrence of* 1 *(respectively $\bot$) from $\vdash \Gamma$ is the main formula of a unique* 1*-rule (respectively $\bot$-rule) of π.*

Proof. By a straightforward induction on π. □

Lemma 16. *Let π be a cut-*free *proof of a sequent $\vdash \Gamma$, where each formula of Γ is* distributed *and with $\Gamma \neq 1$. Then each* 1*-rule in π belongs to a $1/\bot$-pattern.*

Proof. Let r be a 1-rule in π. There is a rule s below r since $\Gamma \neq 1$. This rule s cannot be an *ax*- a $\parr$- nor a 1-rule as one of its hypothesis sequents is $\vdash 1$. Nor can it be a $\otimes$-rule for otherwise its conclusion sequent is of the form $\vdash 1 \otimes A, \Delta$ or $\vdash A \otimes 1, \Delta$, and by the sub-formula property Γ contains a non-distributed formula. Thus, s can only be a $\bot$-rule, hence r and s form a $1/\bot$-pattern. □

Lemma 6. *Let A and B be* distributed *formulas such that $\vdash A^\bot, B$ and $\vdash B^\bot, A$ are both provable. If π is a cut-*free *and ax-*expanded *proof of $\vdash A^\bot, B$, then each* 1*-rule and each $\bot$-rule in π belongs to a $1/\bot$-pattern.*

Proof. By hypothesis, there exists some proof ϕ of $\vdash B^\bot, A$, that we can assume to be *cut*-free and *ax*-expanded. Using Lemma 16, each 1-rule in π or in ϕ belongs to a $1/\bot$-pattern. Hence, by Fact 5, there are more 1 sub-formulas than $\bot$ sub-formulas in $\vdash A^\bot, B$, and similarly in $\vdash B^\bot, A$. However, the number of 1 (respectively $\bot$) sub-formulas of $\vdash A^\bot, B$ is equal to the number of $\bot$ (respectively 1) sub-formulas in $\vdash B^\bot, A$. Therefore, there are as many 1 sub-formulas in $\vdash A^\bot, B$ as there are $\bot$ sub-formulas inside, so again by Fact 5 there are as many 1-rules in π as there as $\bot$-rules. Since a $1/\bot$-pattern contains one 1-rule and one $\bot$-rule, and as each 1-rule belongs to such a pattern, necessarily each $\bot$-rule also belongs to a $1/\bot$-pattern. □

Theorem 8. *Let π and ϕ be cut-*free and *ax-*expanded *proofs of a same sequent $\vdash A^\bot, A$ for A a* distributed *formula. Then $\pi =_{\beta\eta} \phi$ if and only if:*

- *the ax-rules of π and ϕ regroup the exact same occurrences of atoms; and*
- *the $1/\bot$-patterns of π and ϕ regroup the exact same occurrences of* 1 *and $\bot$.*

Proof. This basically is a corollary of known results in the literature. To begin with, $\pi =_{\beta\eta} \phi$ if and only if π and ϕ are linked by a sequence of rules commutations using [5, Theorem 2.50].

If π and ϕ are linked by a sequence of rules commutations, then the two items of the theorem hold because *ax*-rules stay unchanged by rules commutations in multiplicative linear logic, and because no commutation can be applied on neither 1- nor $\bot$-rules by Lemma 6.

If the two items of the theorem hold, then consider a transformation σ that replaces in A (and in $A^\bot$) each 1 by X and each $\bot$ by $X^\bot$, for a fixed atom X. Then, σ also acts on π and ϕ by replacing each $1/\bot$-pattern by $\dfrac{}{\vdash X^\bot, X}\,{}^{(ax)}$. By hypotheses, it follows that the *ax*-rules of $\sigma(\pi)$ and $\sigma(\phi)$ are exactly on the same occurrences. This obviously means that they have the same proof-net (in unit-free multiplicative linear logic) [15], hence $\sigma(\pi)$ and $\sigma(\phi)$ are linked by a sequence of rules commutations using [18, Theorem 1]. Since *ax*-rules do not commute, this sequence of rules commutations is also one between π and ϕ. □

A.2 Finding Rules in Normal Forms

Fact 11. *Let π be a proof of a sequent $\vdash \Gamma$.*

- *Consider in π ax-rules* $\dfrac{}{\vdash X_1^\bot, X_2}\,{}^{(ax)}, \dfrac{}{\vdash X_2^\bot, X_3}\,{}^{(ax)}, \ldots, \dfrac{}{\vdash X_{n-1}^\bot, X_n}\,{}^{(ax)}$ *where $X_1^\bot$ and X_n occur in Γ whereas $X_i^\bot$ and X_i are dual occurrences of a formula introduced by a cut-rule for all $1 < i < n$. There is an ax-rule* $\dfrac{}{\vdash X_1^\bot, X_n}\,{}^{(ax)}$ *in all cut-free ϕ such that π reduces by cut-elimination to ϕ.*
- *Consider in π $1/\bot$-patterns* $\dfrac{\dfrac{}{\vdash 1_2}\,{}^{(1)}}{\vdash \bot_1, 1_2}\,{}^{(\bot)}, \dfrac{\dfrac{}{\vdash 1_3}\,{}^{(1)}}{\vdash \bot_2, 1_3}\,{}^{(\bot)}, \ldots, \dfrac{\dfrac{}{\vdash 1_n}\,{}^{(1)}}{\vdash \bot_{n-1}, 1_n}\,{}^{(\bot)}$ *where $\bot_1$ and 1_n occur in Γ whereas $\bot_i$ and 1_i are dual occurrences of a formula introduced by a cut-rule in π for all $1 < i < n$. There is a $1/\bot$-pattern* $\dfrac{\dfrac{}{\vdash 1_n}\,{}^{(1)}}{\vdash \bot_1, 1_n}\,{}^{(\bot)}$ *in some cut-free ϕ such that π reduces by cut-elimination to ϕ.*

Proof.

- By induction on the reduction sequence $\pi \to^* \phi$. The only cut-elimination step modifying some *ax*-rule is the *ax* key case:

$$\dfrac{\overset{\tau}{\vdash A, \Delta} \qquad \dfrac{}{\vdash A^\bot, A}\,{}^{(ax)}}{\vdash A, \Delta}\,{}^{(cut)} \quad \to \quad \overset{\tau}{\vdash A, \Delta}$$

  This step may delete a $\dfrac{}{\vdash X_{i-1}^\bot, X_i}\,{}^{(ax)}$ and the occurrence X_i (respectively $X_{i-1}^\bot$), so that the *ax*-rule using the occurrence $X_i^\bot$ (respectively X_{i-1}) now use $X_{i-1}^\bot$(respectively X_i) instead. Since in the *cut*-free ϕ only occurrences of Γ are present, we must end with $\dfrac{}{\vdash X_1^\bot, X_n}\,{}^{(ax)}$.

– We apply to π a substitution replacing each $\bot_i$ in these patterns by $X_i^\bot$, each 1_i by X_i, and replace a $1/\bot$-pattern by an ax-rule. This yields a proof π', to which we apply the first item on an arbitrary normal form ϕ': we have $\pi' \to^* \phi'$ with in ϕ' an ax-rule $\dfrac{}{\vdash X_1^\bot, X_n}\,{}^{(ax)}$. Substituting back the introduced $X^\bot$ and X by respectively $\bot$ and 1 in this reduction sequence, we find a proof ϕ such that $\pi \to^* \phi$ – because it is possible to mimic the lacking ax key cases by a $\bot - cut$ commutative case followed by a $\bot - 1$ key case. In particular, we have in ϕ the wanted $1/\bot$-pattern. □

B Proof-Relevant Interpolation with Few Units

We recall and prove here Theorem 2, that is the variant of [21, Theorem 5] and [10, Theorem 4] for proof-relevant interpolation that we use to split idempotents. This result is very similar to the two cited theorems, broadly following the same ideas with just some additional care when handing units so as to get additional properties on their placements in the interpolating proofs. We stated it in second-order linear logic to make these correspondences even more apparent.

Theorem 2. *Let τ be a proof of a sequent $\vdash A^\bot, B$ in second-order linear logic (or in any of its fragments). There exists (in this fragment) a formula C and proofs π and ϕ of respectively $\vdash A^\bot, C$ and $\vdash C^\bot, B$ such that:*

(1) $\dfrac{\overset{\pi}{\vdash A^\bot, C} \qquad \overset{\phi}{\vdash C^\bot, B}}{\vdash A^\bot, B}\,{}_{(cut)}$ *reduces by cut-elimination to τ;*
(2) *if τ is cut-free, then so are π and ϕ;*
(3) *if τ is ax-expanded, then so are π and ϕ;*
(4) *each ax-rule in π (respectively ϕ) uses at least one sub-formula occurrence from $A^\bot$ (respectively from B);*
(5) *each 1-rule in π (respectively ϕ) whose main formula is a 1 from C (respectively from $C^\bot$) is directly above a $\bot$-rule whose main formula is a $\bot$ from $A^\bot$ (respectively from B);*
(6) *each 0 from C (respectively from $C^\bot$) is in π (respectively in ϕ) in the context of a $\top$-rule whose main formula is a $\top$ from $A^\bot$ (respectively from B).*

To prove this theorem, our key tool is to put a proof under a special shape where interpolation is easy, shape that we call *separated*. This uses the notion of *ancestors* of formulas in a proof, see *e.g.* [21, Definition 2]. As an example, in a rule $\dfrac{\vdash A, B, \Gamma}{\vdash A \parr B, \Gamma}\,{}_{(\parr)}$ the (occurrences of) formulas A and B are ancestors of the occurrence) $A \parr B$.

Definition 17. *A proof π of $\vdash \Gamma$ is **separated** if it is equipped with a bipartition $\{\mathcal{S}_1, \mathcal{S}_2\}$ of $\Gamma \uplus \biguplus_i (A_i^\bot \uplus A_i)$ as well as their ancestors in the derivation π, where $(A_i^\bot, A_i)_i$ are the cut-formulas in π, such that:*

- *the bipartition respects the ancestor relation: if B is an ancestor of C, then B and C belong to the same part $\mathcal{S}_i$;*
- *all formulas in the conclusion sequent of each ax-rule are either all in $\mathcal{S}_1$ or all in $\mathcal{S}_2$; and similarly for each $\top$- and !-rule;*
- *for each $\bot$-rule $\dfrac{\vdash \Delta}{\vdash \bot, \Delta}\ (\bot)$ whose principal formula is in $\mathcal{S}_i$, at least one formula of Δ is in $\mathcal{S}_i$; and similarly for each $?_w$-rule;*
- *given an atom X introduced by a $\forall$-rule whose principal formula is in $\mathcal{S}_i$, X cannot appear in any witness of an $\exists$-rule whose principal formula is in $\mathcal{S}_{1-i}$, nor can it appear in a cut-formula of $\mathcal{S}_{1-i}$.*

Remark 18. There is an intuition for Definition 17 in the proof-net syntax of linear logic: it means that each top-most node (with no input) has its endpoints in the same part, and that each jump edge (related to a $\bot$, $?w$ or $\forall$ node) has its endpoints in the same part.

Fact 19. *Observe that a sub-proof π of a separated proof ϕ is also separated, with as bipartition the bipartition of ϕ restricted to the formulas in π.*

Our plan to prove Theorem 2 is the following. First, we ascertain in Appendix B.1 that a separated proof can be interpolated as wanted. Second, we show in Appendix B.2 how to *cut*-expand a given proof into a separated one, provided it is possible to interpolate a separated proof.

B.1 Interpolating a Separated Proof

Proposition 20. *Let π be a separated proof with bipartition $\{\mathcal{S}_1, \mathcal{S}_2\}$ and conclusion sequent $\vdash \Gamma_1, \Gamma_2$, where $\Gamma_1 \subseteq \mathcal{S}_1$ and $\Gamma_2 \subseteq \mathcal{S}_2$. If neither Γ_1 nor Γ_2 is empty, then there exists a formula C and proofs ϕ_1 of $\vdash \Gamma_1, C^\bot$ and ϕ_2 of $\vdash C, \Gamma_2$ such that:*

- $\dfrac{\begin{matrix}\phi_1 \\ \vdash \Gamma_1, C^\bot\end{matrix} \qquad \begin{matrix}\phi_2 \\ \vdash C, \Gamma_2\end{matrix}}{\vdash \Gamma_1, \Gamma_2}\ (cut)$ *reduces by cut-elimination to π without using any ax, $\bot - 1$, exponential nor quantifier key case nor any $\top$ commutative case;*
- *C is obtained by adding binary connectives between all the cut-formulas in $\{A \in \mathcal{S}_2 \mid A$ is a cut-formula of π whose dual occurrence belongs to $\mathcal{S}_1\}$;*
- *cut-rules in ϕ_1 are exactly those of π with both cut-formulas in $\mathcal{S}_1$; and symmetrically for ϕ_2;*
- *ax- 1- $\top$- and !-rules in ϕ_1 and in ϕ_2 are exactly those of π, with the same multiplicity.*

Besides, this result holds in every fragment of second-order linear logic.

Proof. We proceed by structural induction on π, making implicit use of Fact 19 in each case. Most of the wanted properties will follow by construction. We assume *w.l.o.g.* that the principal formula (if any) of the last rule of π is in $\mathcal{S}_1$.

- Assume the last rule of π is a ax- 1- $\top$- or !-rule.
  Since π is separated, this implies that Γ_1 or Γ_2 is empty: contradiction.
- Assume the last rule of π is a $\bot$-rule.
  Then $\pi = \dfrac{\overset{\pi'}{\vdash \Delta_1, \Gamma_2}}{\vdash \bot, \Delta_1, \Gamma_2}\ (\bot)$ with $\Gamma_1 = \bot, \Delta_1$. Since π is separated, Δ_1 is not empty. By induction hypothesis on the separated π', we obtain a formula C' and proofs ϕ'_1 of $\vdash \Delta_1, C'^{\perp}$ and ϕ'_2 of $\vdash C', \Gamma_2$. Set $C \overset{\text{def}}{=} C'$, $\phi_1 \overset{\text{def}}{=} \dfrac{\overset{\phi'_1}{\vdash \Delta_1, C'^{\perp}}}{\vdash \bot, \Delta_1, C'^{\perp}}\ (\bot)$, $\phi_2 \overset{\text{def}}{=} \phi'_2$. The result follows, with as reduction sequence a $\bot - cut$ commutative step then the induction hypothesis:
  $$\dfrac{\dfrac{\overset{\phi'_1}{\vdash \Delta_1, C'^{\perp}}}{\vdash \bot, \Delta_1, C'^{\perp}}\ (\bot) \qquad \overset{\phi'_2}{\vdash C', \Gamma_2}}{\vdash \bot, \Delta_1, \Gamma_2}\ (cut) \to \dfrac{\overset{\phi'_1}{\vdash \Delta_1, C'^{\perp}} \qquad \overset{\phi'_2}{\vdash C', \Gamma_2}}{\dfrac{\vdash \Delta_1, \Gamma_2}{\vdash \bot, \Delta_1, \Gamma_2}\ (\bot)}\ (cut) \to^* \dfrac{\overset{\pi'}{\vdash \Delta_1, \Gamma_2}}{\vdash \bot, \Delta_1, \Gamma_2}\ (\bot)$$
- Assume the last rule of π is a $?w$- or $\forall$-rule.
  The case of a $?w$-rule is similar to the previous case of a $\bot$-rule. The case of a $\forall$-rule also is, provided some care is taken for the introduced variable by using that π is separated.
- Assume the last rule of π is a $\parr$-rule.
  Then $\pi = \dfrac{\overset{\pi'}{\vdash A, B, \Delta_1, \Gamma_2}}{\vdash A \parr B, \Delta_1, \Gamma_2}\ (\parr)$ with $\Gamma_1 = A \parr B, \Delta_1$. By induction hypothesis on the separated π', we obtain a formula C' and proofs ϕ'_1 of $\vdash A, B, \Delta_1, C'^{\perp}$ and ϕ'_2 of $\vdash C', \Gamma_2$. Set $C \overset{\text{def}}{=} C'$, $\phi_1 \overset{\text{def}}{=} \dfrac{\overset{\phi'_1}{\vdash A, B, \Delta_1, C'^{\perp}}}{\vdash A \parr B, \Delta_1, C'^{\perp}}\ (\parr)$, $\phi_2 \overset{\text{def}}{=} \phi'_2$. The result follows, with as reduction sequence a $\parr - cut$ commutative step then the induction hypothesis:
  $$\dfrac{\dfrac{\overset{\phi'_1}{\vdash A, B, \Delta_1, C'^{\perp}}}{\vdash A \parr B, \Delta_1, C'^{\perp}}\ (\parr) \qquad \overset{\phi'_2}{\vdash C', \Gamma_2}}{\vdash A \parr B, \Delta_1, \Gamma_2}\ (cut) \to \dfrac{\overset{\phi'_1}{\vdash A, B, \Delta_1, C'^{\perp}} \qquad \overset{\phi'_2}{\vdash C', \Gamma_2}}{\dfrac{\vdash A, B, \Delta_1, \Gamma_2}{\vdash A \parr B, \Delta_1, \Gamma_2}\ (\parr)}\ (cut) \to^* \dfrac{\overset{\pi'}{\vdash A, B, \Delta_1, \Gamma_2}}{\vdash A \parr B, \Delta_1, \Gamma_2}\ (\parr)$$
- Assume the last rule of π is a $\oplus_1$- $\oplus_2$- $?d$- $?c$- or $\exists$-rule.
  These cases are all similar to the previous case of a $\parr$-rule.
- Assume the last rule of π is a $\otimes$-rule.
  Then $\pi = \dfrac{\overset{\pi^l}{\vdash A, \Gamma_1^l, \Gamma_2^l} \qquad \overset{\pi^r}{\vdash B, \Gamma_1^r, \Gamma_2^r}}{\vdash A \otimes B, \Gamma_1^l, \Gamma_1^r, \Gamma_2^l, \Gamma_2^r}\ (\otimes)$ with $\Gamma_1 = A \otimes B, \Gamma_1^l, \Gamma_1^r$ and $\Gamma_2 = \Gamma_2^l, \Gamma_2^r$. Since Γ_2 is not empty, at least one of Γ_2^l and Γ_2^r is not empty: say by symmetry Γ_2^l is not empty. By induction hypothesis on the separated π^l, we obtain a formula C^l and proofs ϕ_1^l of $\vdash A, \Gamma_1^l, C^{l\perp}$ and ϕ_2^l of $\vdash C^l, \Gamma_2^l$. We distinguish cases according to the emptiness of Γ_2^r.

- Assume Γ_2^r is not empty.
  By induction hypothesis on the separated π^r, we obtain a formula C^r and proofs ϕ_1^r of $\vdash B, \Gamma_1^r, C^{r\perp}$ and ϕ_2^r of $\vdash C^r, \Gamma_2^r$. Set $C \stackrel{\text{def}}{=} C^l \otimes C^r$,
  $\phi_1 \stackrel{\text{def}}{=} \dfrac{\dfrac{\phi_1^l \quad \phi_1^r}{\vdash A, \Gamma_1^l, C^{l\perp} \quad \vdash B, \Gamma_1^r, C^{r\perp}}\,(\otimes)}{\dfrac{\vdash A \otimes B, \Gamma_1^l, \Gamma_1^r, C^{l\perp}, C^{r\perp}}{\vdash A \otimes B, \Gamma_1^l, \Gamma_1^r, C^{l\perp} \parr C^{r\perp}}\,(\parr)}$, $\phi_2 \stackrel{\text{def}}{=} \dfrac{\phi_2^l \quad \phi_2^r}{\vdash C^l, \Gamma_2^l \quad \vdash C^r, \Gamma_2^r}\,(\otimes)$.
  The result follows, with as reduction sequence a $\parr - \otimes$ key step, two $\otimes -$ *cut* commutative steps then the induction hypotheses:

$$\dfrac{\dfrac{\dfrac{\phi_1^l \quad \phi_1^r}{\vdash A, \Gamma_1^l, C^{l\perp} \quad \vdash B, \Gamma_1^r, C^{r\perp}}\,(\otimes)}{\vdash A \otimes B, \Gamma_1^l, \Gamma_1^r, C^{l\perp}, C^{r\perp}}\,(\parr) \quad \dfrac{\phi_2^l \quad \phi_2^r}{\vdash C^l, \Gamma_2^l \quad \vdash C^r, \Gamma_2^r}\,(\otimes)}{\vdash A \otimes B, \Gamma_1^l, \Gamma_1^r, C^{l\perp} \parr C^{r\perp} \quad \vdash C^l \otimes C^r, \Gamma_2^l, \Gamma_2^r}\,(cut) \;\to\; \dfrac{\dfrac{\dfrac{\phi_1^l \quad \phi_1^r}{\vdash A, \Gamma_1^l, C^{l\perp} \quad \vdash B, \Gamma_1^r, C^{r\perp}}\,(\otimes) \quad \phi_2^l}{\vdash A \otimes B, \Gamma_1^l, \Gamma_1^r, C^{l\perp}, C^{r\perp} \quad \vdash C^l, \Gamma_2^l}\,(cut) \quad \phi_2^r}{\vdash A \otimes B, \Gamma_1^l, \Gamma_1^r, \Gamma_2^l, C^{r\perp} \quad \vdash C^r, \Gamma_2^r}\,(cut)}{\vdash A \otimes B, \Gamma_1^l, \Gamma_1^r, \Gamma_2^l, \Gamma_2^r}$$

$$\to \dfrac{\dfrac{\phi_1^l \quad \phi_2^l}{\vdash A, \Gamma_1^l, C^{l\perp} \quad \vdash C^l, \Gamma_2^l}\,(cut) \quad \dfrac{\phi_1^r}{\vdash B, \Gamma_1^r, C^{r\perp}}\,(\otimes) \quad \phi_2^r}{\vdash A, \Gamma_1^l, \Gamma_1^r, \Gamma_2^l, C^{r\perp} \quad \vdash C^r, \Gamma_2^r}\,(cut)}{\vdash A \otimes B, \Gamma_1^l, \Gamma_1^r, \Gamma_2^l, \Gamma_2^r}$$

$$\to \dfrac{\dfrac{\phi_1^l \quad \phi_2^l}{\vdash A, \Gamma_1^l, C^{l\perp} \quad \vdash C^l, \Gamma_2^l}\,(cut) \quad \dfrac{\phi_1^r \quad \phi_2^r}{\vdash B, \Gamma_1^r, C^{r\perp} \quad \vdash C^r, \Gamma_2^r}\,(cut)}{\vdash A, \Gamma_1^l, \Gamma_1^r, \Gamma_2^l, \Gamma_2^r}\,(\otimes) \to^* \dfrac{\dfrac{\pi^l}{\vdash A, \Gamma_1^l, \Gamma_2^l} \quad \dfrac{\pi^r}{\vdash B, \Gamma_1^r, \Gamma_2^r}}{\vdash A \otimes B, \Gamma_1^l, \Gamma_1^r, \Gamma_2^l, \Gamma_2^r}\,(\otimes)$$

- Assume Γ_2^r is empty.
  Set $C \stackrel{\text{def}}{=} C^l$, $\phi_1 \stackrel{\text{def}}{=} \dfrac{\phi_1^l \quad \pi^r}{\vdash A, \Gamma_1^l, C^{l\perp} \quad \vdash B, \Gamma_1^r}\,(\otimes)$, $\phi_2 \stackrel{\text{def}}{=} \phi_2^l$. The result follows, with as reduction sequence a $\otimes -$ *cut* commutative step then the induction hypothesis:

$$\dfrac{\dfrac{\phi_1^l \quad \pi^r}{\vdash A, \Gamma_1^l, C^{l\perp} \quad \vdash B, \Gamma_1^r}\,(\otimes) \quad \phi_2^l}{\vdash A \otimes B, \Gamma_1^l, \Gamma_1^r, C^{l\perp} \quad \vdash C^l, \Gamma_2^l}\,(cut) \;\to\; \dfrac{\dfrac{\phi_1^l \quad \phi_2^l}{\vdash A, \Gamma_1^l, C^{l\perp} \quad \vdash C^l, \Gamma_2^l}\,(cut) \quad \pi^r}{\vdash A, \Gamma_1^l, \Gamma_2^l \quad \vdash B, \Gamma_1^r}\,(\otimes) \;\to^* \dfrac{\dfrac{\pi^l}{\vdash A, \Gamma_1^l, \Gamma_2^l} \quad \dfrac{\pi^r}{\vdash B, \Gamma_1^r}}{\vdash A \otimes B, \Gamma_1^l, \Gamma_1^r, \Gamma_2^l}\,(\otimes)$$

– Assume the last rule of π is a *cut*-rule.
  If its *cut*-formulas $A^\perp$ and A both belong to $\mathcal{S}_1$, or both belong to $\mathcal{S}_2$, we can proceed similarly as in the previous case of a $\otimes$-rule. Hence, assume by symmetry that $A^\perp \in \mathcal{S}_1$ and $A \in \mathcal{S}_2$. Then $\pi = \dfrac{\pi^l \quad \pi^r}{\vdash A^\perp, \Gamma_1^l, \Gamma_2^l \quad \vdash \Gamma_1^r, A, \Gamma_2^r}\,(cut)$
  with $\Gamma_1 = \Gamma_1^l, \Gamma_1^r$ and $\Gamma_2 = \Gamma_2^l, \Gamma_2^r$. There are four sub-cases according to the emptiness of Γ_2^l and Γ_1^r. If both are empty, then π is already of the required form. By symmetry, there are two cases left: both are non-empty, or Γ_2^l is not empty and Γ_1^r is empty. In both cases, by induction hypothesis on the separated π^l, we obtain a formula C^l and proofs ϕ_1^l of $\vdash A^\perp, \Gamma_1^l, C^{l\perp}$ and ϕ_2^l of $\vdash C^l, \Gamma_2^l$.
  - Assume Γ_2^l and Γ_1^r are not empty.
    By induction hypothesis on the separated π^r, we obtain a formula C^r and

proofs ϕ_1^r of $\vdash \Gamma_1^r, C^{r\perp}$ and ϕ_2^r of $\vdash C^r, A, \Gamma_2^r$. Set $C \stackrel{\text{def}}{=} C^l \otimes (A \parr C^r)$[3],

$$\phi_1 \stackrel{\text{def}}{=} \dfrac{\dfrac{\phi_1^l \quad \phi_1^r}{\vdash A^\perp, \Gamma_1^l, C^{l\perp} \quad \vdash \Gamma_1^r, C^{r\perp}}\,(\otimes)}{\vdash C^{l\perp}, A^\perp \otimes C^{r\perp}, \Gamma_1^l, \Gamma_1^r}\,(\parr)}{\vdash C^{l\perp} \parr (A^\perp \otimes C^{r\perp}), \Gamma_1^l, \Gamma_1^r}, \quad \phi_2 \stackrel{\text{def}}{=} \dfrac{\phi_2^l \quad \dfrac{\phi_2^r}{\vdash C^r, A, \Gamma_2^r}\,(\parr)}{\vdash C^l, \Gamma_2^l \quad \vdash A \parr C^r, \Gamma_2^r}\,(\otimes)}{\vdash C^l \otimes (A \parr C^r), \Gamma_2^l, \Gamma_2^r}.$$

The result follows, with as reduction sequence two $\parr - \otimes$ key steps, a $cut - cut$ commutative step then the induction hypotheses:

$$\dfrac{\dfrac{\dfrac{\dfrac{\phi_1^l}{\vdash A^\perp, \Gamma_1^l, C^{l\perp}} \quad \dfrac{\phi_1^r}{\vdash \Gamma_1^r, C^{r\perp}}}{\vdash C^{l\perp}, A^\perp \otimes C^{r\perp}, \Gamma_1^l, \Gamma_1^r}}{\vdash C^{l\perp} \parr (A^\perp \otimes C^{r\perp}), \Gamma_1^l, \Gamma_1^r} \quad \dfrac{\phi_2^l \quad \dfrac{\dfrac{\phi_2^r}{\vdash C^r, A, \Gamma_2^r}}{\vdash A \parr C^r, \Gamma_2^r}}{\vdash C^l \otimes (A \parr C^r), \Gamma_2^l, \Gamma_2^r}}{\vdash \Gamma_1^l, \Gamma_1^r, \Gamma_2^l, \Gamma_2^r} \to \dfrac{\dfrac{\dfrac{\phi_1^l}{\vdash A^\perp, \Gamma_1^l, C^{l\perp}} \quad \dfrac{\phi_1^r}{\vdash \Gamma_1^r, C^{r\perp}}}{\vdash C^{l\perp}, A^\perp \otimes C^{r\perp}, \Gamma_1^l, \Gamma_1^r} \quad \dfrac{\dfrac{\phi_2^r}{\vdash C^r, A, \Gamma_2^r}}{\vdash A \parr C^r, \Gamma_2^r}}{\vdash C^{l\perp}, \Gamma_1^l, \Gamma_1^r, \Gamma_2^r} \quad \phi_2^l}{\vdash \Gamma_1^l, \Gamma_1^r, \Gamma_2^l, \Gamma_2^r}$$

$$\to \dfrac{\dfrac{\phi_1^l}{\vdash A^\perp, \Gamma_1^l, C^{l\perp}} \quad \dfrac{\dfrac{\phi_1^r}{\vdash \Gamma_1^r, C^{r\perp}} \quad \dfrac{\phi_2^r}{\vdash C^r, A, \Gamma_2^r}}{\vdash \Gamma_1^r, A, \Gamma_2^r}}{\vdash C^{l\perp}, \Gamma_1^l, \Gamma_1^r, \Gamma_2^r} \quad \phi_2^l}{\vdash \Gamma_1^l, \Gamma_1^r, \Gamma_2^l, \Gamma_2^r}$$

$$\to \dfrac{\dfrac{\phi_1^l}{\vdash A^\perp, \Gamma_1^l, C^{l\perp}} \quad \dfrac{\phi_2^l}{\vdash C^l, \Gamma_2^l}}{\vdash A^\perp, \Gamma_1^l, \Gamma_2^l} \quad \dfrac{\dfrac{\phi_1^r}{\vdash \Gamma_1^r, C^{r\perp}} \quad \dfrac{\phi_2^r}{\vdash C^r, A, \Gamma_2^r}}{\vdash \Gamma_1^r, A, \Gamma_2^r}}{\vdash \Gamma_1^l, \Gamma_1^r, \Gamma_2^l, \Gamma_2^r} \to^* \dfrac{\dfrac{\pi^l}{\vdash A^\perp, \Gamma_1^l, \Gamma_2^l} \quad \dfrac{\pi^r}{\vdash \Gamma_1^r, A, \Gamma_2^r}}{\vdash \Gamma_1^l, \Gamma_1^r, \Gamma_2^l, \Gamma_2^r}$$

- Assume Γ_2^l is not empty and Γ_1^r is empty.

  Set $C \stackrel{\text{def}}{=} C^l \parr A$, $\phi_1 \stackrel{\text{def}}{=} \dfrac{\dfrac{\phi_1^l}{\vdash A^\perp, \Gamma_1^l, C^{l\perp}}\,(\parr)}{\vdash C^{l\perp} \parr A^\perp, \Gamma_1^l}$, $\phi_2 \stackrel{\text{def}}{=} \dfrac{\phi_2^l \quad \dfrac{\pi^r}{\vdash A, \Gamma_2^r}}{\vdash C^l, \Gamma_2^l \quad \vdash A, \Gamma_2^r}\,(\otimes)}{\vdash C^l \otimes A, \Gamma_2^l, \Gamma_2^r}$.

  The result follows, with as reduction sequence a $\parr - \otimes$ key step then the induction hypothesis:

$$\dfrac{\dfrac{\dfrac{\phi_1^l}{\vdash A^\perp, \Gamma_1^l, C^{l\perp}}\,(\parr)}{\vdash C^{l\perp} \parr A^\perp, \Gamma_1^l} \quad \dfrac{\phi_2^l \quad \dfrac{\pi^r}{\vdash A, \Gamma_2^r}}{\vdash C^l, \Gamma_2^l \quad \vdash A, \Gamma_2^r}\,(\otimes)}{\vdash C^l \otimes A, \Gamma_2^l, \Gamma_2^r}\,(cut) \to \dfrac{\dfrac{\phi_1^l}{\vdash A^\perp, \Gamma_1^l, C^{l\perp}} \quad \dfrac{\phi_2^l}{\vdash C^l, \Gamma_2^l}\,(cut) \quad \dfrac{\pi^r}{\vdash A, \Gamma_2^r}\,(cut)}{\vdash \Gamma_1^l, \Gamma_2^l, \Gamma_2^r} \to^* \dfrac{\dfrac{\pi^l}{\vdash A^\perp, \Gamma_1^l, \Gamma_2^l} \quad \dfrac{\pi^r}{\vdash A, \Gamma_2^r}\,(cut)}{\vdash \Gamma_1^l, \Gamma_2^l, \Gamma_2^r}$$

– Assume the last rule of π is a &-rule.

Then $\pi = \dfrac{\pi^l \quad \pi^r}{\vdash A, \Delta_1, \Gamma_2 \quad \vdash B, \Delta_1, \Gamma_2}\,(\&)$ with $\Gamma_1 = A \& B, \Delta_1$. By induction hypothesis on the separated π^l, we obtain a formula C^l and proofs ϕ_1^l of $\vdash A, \Delta_1, C^{l\perp}$ and ϕ_2^l of $\vdash C^l, \Gamma_2$. By induction hypothesis on the separated π^r, we obtain a formula C^r and proofs ϕ_1^r of $\vdash B, \Delta_1, C^{r\perp}$ and ϕ_2^r of $\vdash C^r, \Gamma_2$. Set

$$C \stackrel{\text{def}}{=} C^l \,\&\, C^r,\ \phi_1 \stackrel{\text{def}}{=} \dfrac{\dfrac{\phi_1^l}{\vdash A, \Delta_1, C^{l\perp}}\,(\oplus_1) \qquad \dfrac{\phi_1^r}{\vdash B, \Delta_1, C^{r\perp}}\,(\oplus_2)}{\vdash A \,\&\, B, \Delta_1, C^{l\perp} \oplus C^{r\perp}}\,(\&),$$

[3] Other choices for C are possible, such as $C \stackrel{\text{def}}{=} (C^l \otimes A) \parr C^r$.

$\phi_2 \stackrel{\text{def}}{=} \dfrac{\overset{\phi_2^l}{\vdash C^l, \Gamma_2} \quad \overset{\phi_2^r}{\vdash C^r, \Gamma_2}}{\vdash C^l \,\&\, C^r, \Gamma_2}\,(\&)$. The result follows, with as reduction sequence a $\& - \mathit{cut}$ commutative step, two $\& - \oplus$ key cases then the induction hypotheses:

$$\dfrac{\dfrac{\dfrac{\overset{\phi_1^l}{\vdash A, \Delta_1, C^{l\perp}}}{\vdash A, \Delta_1, C^{l\perp} \oplus C^{r\perp}}\,(\oplus_1) \quad \dfrac{\overset{\phi_1^r}{\vdash B, \Delta_1, C^{r\perp}}}{\vdash B, \Delta_1, C^{l\perp} \oplus C^{r\perp}}\,(\oplus_2)}{\vdash A \,\&\, B, \Delta_1, C^{l\perp} \oplus C^{r\perp}}\,(\&) \quad \dfrac{\overset{\phi_2^l}{\vdash C^l, \Gamma_2} \quad \overset{\phi_2^r}{\vdash C^r, \Gamma_2}}{\vdash C^l \,\&\, C^r, \Gamma_2}\,(\&)}{\vdash A \,\&\, B, \Delta_1, \Gamma_2}\,(\mathit{cut})$$

$$\to \dfrac{\dfrac{\overset{\phi_1^l}{\vdash A, \Delta_1, C^{l\perp}}}{\vdash A, \Delta_1, C^{l\perp} \oplus C^{r\perp}}\,(\oplus_1) \quad \dfrac{\overset{\phi_2^l}{\vdash C^l, \Gamma_2} \quad \overset{\phi_2^r}{\vdash C^r, \Gamma_2}}{\vdash C^l \,\&\, C^r, \Gamma_2}\,(\&)}{\vdash A, \Delta_1, \Gamma_2}\,(\mathit{cut}) \quad \dfrac{\dfrac{\overset{\phi_1^r}{\vdash B, \Delta_1, C^{r\perp}}}{\vdash B, \Delta_1, C^{l\perp} \oplus C^{r\perp}}\,(\oplus_2) \quad \dfrac{\overset{\phi_2^l}{\vdash C^l, \Gamma_2} \quad \overset{\phi_2^r}{\vdash C^r, \Gamma_2}}{\vdash C^l \,\&\, C^r, \Gamma_2}\,(\&)}{\vdash B, \Delta_1, \Gamma_2}\,(\mathit{cut})}{\vdash A \,\&\, B, \Delta_1, \Gamma_2}\,(\&)$$

$$\to\to \dfrac{\dfrac{\overset{\phi_1^l}{\vdash A, \Delta_1, C^{l\perp}} \quad \overset{\phi_2^l}{\vdash C^l, \Gamma_2}}{\vdash A, \Delta_1, \Gamma_2}\,(\mathit{cut}) \quad \dfrac{\overset{\phi_1^r}{\vdash B, \Delta_1, C^{r\perp}} \quad \overset{\phi_2^r}{\vdash C^r, \Gamma_2}}{\vdash B, \Delta_1, \Gamma_2}\,(\mathit{cut})}{\vdash A \,\&\, B, \Delta_1, \Gamma_2}\,(\&) \to^* \dfrac{\overset{\pi^l}{\vdash A, \Delta_1, \Gamma_2} \quad \overset{\pi^r}{\vdash B, \Delta_1, \Gamma_2}}{\vdash A \,\&\, B, \Delta_1, \Gamma_2}\,(\&)$$

□

Remark 21. Observe that Proposition 20, when applied on an *ax*-expanded proof π whose *cut*-rules all give dual *cut*-formulas in different parts of the bipartition, yields proofs ϕ_1 and ϕ_2 that are *cut*-free and *ax*-expanded.

B.2 Separating and Interpolating a Proof

We can now directly proceed to prove Theorem 2 via uses of Proposition 20.

Proof of Theorem 2. We assume given a bipartition $(\mathcal{S}_1, \mathcal{S}_2)$ of the conclusion sequent of τ and of its *cut*-formulas (and expanded to their ancestors) such that $A^\perp$ belongs to $\mathcal{S}_1$ and B belongs to $\mathcal{S}_2$ – in other words, our construction works for any choices of parts for the *cut*-formulas. We will build a separated proof τ' that reduces to τ by *cut*-elimination and whose bipartition $(\mathcal{S}'_1, \mathcal{S}'_2)$ respects $\mathcal{S}_1 \subseteq \mathcal{S}'_1$ and $\mathcal{S}_2 \subseteq \mathcal{S}'_2$. Once it is done, we directly conclude by applying Proposition 20 on τ', the wanted properties holding by construction (*c.f.* Remark 21).

The main difficulty is that to build τ' we may need to interpolate some of the sub-proofs of τ. Henceforth, we proceed by induction on the number of rules in τ not respecting the definition of separated (Definition 17). If there is no such rule, then we are done. Otherwise, take any of these rules and consider its kind. We recall indices are used to identify occurrences.

– Assume there is an *ax*-rule $\dfrac{}{\vdash D_1^\perp, D_2}\,(ax)$ in τ such that, by symmetry, $D_1^\perp \in \mathcal{S}_1$ and $D_2 \in \mathcal{S}_2$. Set τ' as τ where this rule is replaced by:

$$\dfrac{\dfrac{}{\vdash D_1^\perp, D_3}\,(ax) \quad \dfrac{}{\vdash D_4^\perp, D_2}\,(ax)}{\vdash D_1^\perp, D_2}\,(\mathit{cut})$$

and $\mathcal{S}'_1 \stackrel{\text{def}}{=} \mathcal{S}_1 \uplus \{D_3\}$ and $\mathcal{S}'_2 \stackrel{\text{def}}{=} \mathcal{S}_2 \uplus \{D_4^\perp\}$. Hence, $\tau' \to \tau$ using an *ax* key case and we conclude by induction hypothesis on τ' with bipartition $(\mathcal{S}'_1, \mathcal{S}'_2)$.

– Assume there is a $\top$-rule $\dfrac{}{\vdash \top_1, \Gamma, \Delta}\ (\top)$ in τ such that, by symmetry, $\top \in \mathcal{S}_1$, $\Gamma \subseteq \mathcal{S}_1$ and $\emptyset \neq \Delta \subseteq \mathcal{S}_2$. Set τ' as τ where this rule is replaced by:

$$\dfrac{\dfrac{}{\vdash \top_1, \Gamma, 0_3}\ (\top) \qquad \dfrac{}{\vdash \top_4, \Delta}\ (\top)}{\vdash \top_1, \Gamma, \Delta}\ (cut)$$

and $\mathcal{S}'_1 \stackrel{\text{def}}{=} \mathcal{S}_1 \uplus \{0_3\}$ and $\mathcal{S}'_2 \stackrel{\text{def}}{=} \mathcal{S}_2 \uplus \{\top_4\}$. Hence, $\tau' \to \tau$ using a $\top - cut$ commutative case and we conclude by induction hypothesis on τ' with bipartition $(\mathcal{S}'_1, \mathcal{S}'_2)$.

– Assume there is a !-rule $\dfrac{\overset{\pi}{\vdash D, ?\Gamma, ?\Delta}}{\vdash !D, ?\Gamma, ?\Delta}\ (!)$ in τ such that, by symmetry, $!D \in \mathcal{S}_1$, $?\Gamma \subseteq \mathcal{S}_1$ and $\emptyset \neq ?\Delta \subseteq \mathcal{S}_2$. By induction hypothesis on π, there exists a separated π' that reduces to it, with bipartition $(\mathcal{S}'_1, \mathcal{S}'_2)$. By an application of Proposition 20 on π', we obtain a formula C and proofs ϕ_1 and ϕ_2. Set τ' as τ where this !-rule and the sub-proof π are replaced by:

$$\dfrac{\dfrac{\dfrac{\overset{\phi_1}{\vdash D, ?\Gamma, C^\perp}}{\vdash D, ?\Gamma, ?C^\perp}\ (?d)}{\vdash !D, ?\Gamma, ?C^\perp}\ (!) \qquad \dfrac{\overset{\phi_2}{\vdash C, ?\Delta}}{\vdash !C, ?\Delta}\ (!)}{\vdash !D, ?\Gamma, ?\Delta}\ (cut)$$

and $\mathcal{S}''_1 \stackrel{\text{def}}{=} \mathcal{S}'_1 \uplus \{?C^\perp\}$ and $\mathcal{S}''_2 \stackrel{\text{def}}{=} \mathcal{S}'_2 \uplus \{!C\}$. Observe that $\tau' \to^* \tau$ using a $! - cut$ commutative case, a $?d - !$ key case then the reduction sequence given by the application of Proposition 20 on π'. Therefore, we can now conclude by induction hypothesis on τ' with bipartition $(\mathcal{S}''_1, \mathcal{S}''_2)$.

– Assume there is a $\perp$-rule $\dfrac{\overset{\pi}{\vdash \Gamma}}{\vdash \perp_1, \Gamma}\ (\perp)$ in τ such that, by symmetry, $\perp_1 \in \mathcal{S}_1$ and $\Gamma \subseteq \mathcal{S}_2$. Set τ' as τ where this sub-proof is replaced by:

$$\dfrac{\dfrac{\dfrac{}{\vdash 1_3}\ (1)}{\vdash \perp_1, 1_3}\ (\perp) \qquad \dfrac{\overset{\pi}{\vdash \Gamma}}{\vdash \perp_4, \Gamma}\ (\perp)}{\vdash \perp_1, \Gamma}\ (cut)$$

and $\mathcal{S}'_1 \stackrel{\text{def}}{=} \mathcal{S}_1 \uplus \{1_3\}$ and $\mathcal{S}'_2 \stackrel{\text{def}}{=} \mathcal{S}_2 \uplus \{\perp_4\}$. Hence, $\tau' \to \tau$ using a $\perp - 1$ key case and we conclude by induction hypothesis on τ' with bipartition $(\mathcal{S}'_1, \mathcal{S}'_2)$.

– The case of a $?w$-rule is similar to the previous case of a $\perp$-rule.

– Assume there is a $\forall$-rule $\dfrac{\overset{\pi}{\vdash \Gamma, \Delta}}{\vdash \forall XA, \Gamma, \Delta}\ (\forall)$ in τ such that, by symmetry, $\forall XA \in \mathcal{S}_1$, $\Gamma \subseteq \mathcal{S}_1$ and $\Delta \subseteq \mathcal{S}_2$. By induction hypothesis on π, there exists a separated π' that reduces to it, with bipartition $(\mathcal{S}'_1, \mathcal{S}'_2)$. By an application of Proposition 20 on π', we obtain a formula C and proofs ϕ_1 and

ϕ_2. Take a fresh atom Y and set τ' as τ where this $\forall$-rule and the sub-proof π are replaced by:

$$\dfrac{\dfrac{\dfrac{\dfrac{\phi_1}{\vdash A, \Gamma, C^\perp}}{\vdash A, \Gamma, \exists Y C^\perp[Y/X]}\,(\exists)}{\vdash \forall X A, \Gamma, \exists Y C^\perp[Y/X]}\,(\forall) \qquad \dfrac{\dfrac{\phi_2[Y/X]}{\vdash C[Y/X], \Delta}}{\vdash \forall Y C[Y/X], \Delta}\,(\forall)}{\vdash \forall X A, \Gamma, \Delta}\,(cut)$$

and $\mathcal{S}_1'' \stackrel{\text{def}}{=} \mathcal{S}_1' \uplus \{\exists Y C^\perp[Y/X]\}$ and $\mathcal{S}_2'' \stackrel{\text{def}}{=} \mathcal{S}_2' \uplus \{\forall Y C[Y/X]\}$. Observe that $\tau' \to^* \tau$ using a $\forall - cut$ commutative case, a $\forall - \exists$ key case then the reduction sequence given by the application of Proposition 20 on π'. Therefore, we can now conclude by induction hypothesis on τ' with as bipartition $(\mathcal{S}_1'', \mathcal{S}_2'')$. □

Remark 22. It is possible to transpose this demonstration in sequent calculus to the proof-net syntax of multiplicative-exponential linear logic, following [10,11]. The corresponding methodology is to introduce cuts to get the equivalent of a separated proof, and then to merge cuts by applying the algorithm from [7, Appendix E].

Remark 23. The demonstration of Theorem 2 is constructive and yields an algorithm to interpolate a proof. Its time complexity is upper-bounded by the number of rules preventing τ to be separated times the number $|\tau|$ of rules in τ, so belongs to $\mathcal{O}(|\tau|^2)$ – *i.e.* this is a quadratic algorithm. Meanwhile, the number of atoms and units in the produced interpolation formula is linear in the number of rules preventing τ to be separated plus the size of the *cut*-formulas of τ whose dual belongs to the other part of the bipartition. In particular, when τ is *cut*-free, the size of the interpolant is linear: it belongs to $\mathcal{O}(|\tau|)$.

C Proofs of the "Up To Isomorphism" Claims

This technical section details how in the proof of Theorem 14 in Section 4 we can assume formulas to be distributed, which was necessary to apply Theorem 8. In all this section, we will consider proofs up to cut-elimination and axiom-expansion. Henceforth, we adopt the following convention: all named proofs introduced in this section are normal forms, *i.e.* are *cut*-free and *ax*-expanded.

We consider here a normal idempotent τ of $\vdash A^\perp, A$. Let A' be a distributed formula such that $A \simeq A'$: by definition of an isomorphism, there exists proofs μ and ρ respectively of $\vdash A^\perp, A'$ and $\vdash A'^\perp, A$ such that these two equations hold:

$$\dfrac{\begin{array}{cc}\mu & \rho\\ \vdash A^\perp, A' & \vdash A'^\perp, A\end{array}}{\vdash A^\perp, A}\,(cut) \quad =_{\beta\eta} \quad \dfrac{}{\vdash A^\perp, A}\,(ax) \tag{4}$$

$$\dfrac{\begin{array}{cc}\rho & \mu\\ \vdash A'^\perp, A & \vdash A^\perp, A'\end{array}}{\vdash A'^\perp, A'}\,(cut) \quad =_{\beta\eta} \quad \dfrac{}{\vdash A'^\perp, A'}\,(ax) \tag{5}$$

We define τ' as any normal form of $\dfrac{\dfrac{\rho}{\vdash A'^{\perp}, A} \quad \dfrac{\dfrac{\tau}{\vdash A^{\perp}, A} \quad \dfrac{\mu}{\vdash A^{\perp}, A'}}{\vdash A^{\perp}, A'}\,(cut)}{\vdash A'^{\perp}, A'}\,(cut)$, that can be viewed as "τ up to the isomorphism $A \simeq A'$". In particular, we have:

$$\tau' \quad =_{\beta\eta} \quad \dfrac{\dfrac{\rho}{\vdash A'^{\perp}, A} \quad \dfrac{\dfrac{\tau}{\vdash A^{\perp}, A} \quad \dfrac{\mu}{\vdash A^{\perp}, A'}}{\vdash A^{\perp}, A'}\,(cut)}{\vdash A'^{\perp}, A'}\,(cut) \tag{6}$$

We will show how results about (split) idempotency transfer between τ and τ', those results being used in the proof of Theorem 14.

Lemma 24. *The proof τ' is an idempotent.*

Proof. This follows from a simple computation.

$$\dfrac{\dfrac{\tau'}{\vdash A'^{\perp}, A'} \quad \dfrac{\tau'}{\vdash A'^{\perp}, A'}\,(cut)}{\vdash A'^{\perp}, A'}$$

$$=_{\beta\eta} \dfrac{\dfrac{\rho}{\vdash A'^{\perp}, A} \quad \dfrac{\dfrac{\tau}{\vdash A^{\perp}, A} \quad \dfrac{\mu}{\vdash A^{\perp}, A'}}{\vdash A^{\perp}, A'}\,(cut)}{\vdash A'^{\perp}, A'}\,(cut) \qquad \dfrac{\dfrac{\rho}{\vdash A'^{\perp}, A} \quad \dfrac{\dfrac{\tau}{\vdash A^{\perp}, A} \quad \dfrac{\mu}{\vdash A^{\perp}, A'}}{\vdash A^{\perp}, A'}\,(cut)}{\vdash A'^{\perp}, A'}\,(cut)}{\vdash A'^{\perp}, A'}\,(cut)$$

by Equation (6)

$$=_{\beta\eta} \dfrac{\dfrac{\rho}{\vdash A'^{\perp}, A} \quad \dfrac{\dfrac{\tau}{\vdash A^{\perp}, A} \quad \dfrac{\dfrac{\mu}{\vdash A^{\perp}, A'} \quad \dfrac{\rho}{\vdash A'^{\perp}, A}\,(cut)}{\vdash A^{\perp}, A}\,(cut)}{\vdash A^{\perp}, A}\,(cut) \quad \dfrac{\dfrac{\tau}{\vdash A^{\perp}, A} \quad \dfrac{\mu}{\vdash A^{\perp}, A'}\,(cut)}{\vdash A^{\perp}, A}\,(cut)}{\vdash A'^{\perp}, A'}\,(cut)$$

by several $cut - cut$ commutative cases of cut-elimination

$$=_{\beta\eta} \dfrac{\dfrac{\rho}{\vdash A'^{\perp}, A} \quad \dfrac{\dfrac{\tau}{\vdash A^{\perp}, A} \quad \dfrac{}{\vdash A^{\perp}, A}\,(ax)}{\vdash A^{\perp}, A}\,(cut) \quad \dfrac{\dfrac{\tau}{\vdash A^{\perp}, A} \quad \dfrac{\mu}{\vdash A^{\perp}, A'}\,(cut)}{\vdash A^{\perp}, A}\,(cut)}{\vdash A'^{\perp}, A'}\,(cut)$$

by Equation (4)

$$=_{\beta\eta} \dfrac{\dfrac{\rho}{\vdash A'^{\perp}, A} \qquad \dfrac{\dfrac{\tau}{\vdash A^{\perp}, A} \quad \dfrac{\tau}{\vdash A^{\perp}, A}\ (cut)}{\vdash A^{\perp}, A} \qquad \dfrac{\mu}{\vdash A^{\perp}, A'}\ (cut)}{\vdash A^{\perp}, A'}\ (cut)}{\vdash A'^{\perp}, A'}$$

by an *ax* key case of cut-elimination

$$=_{\beta\eta} \dfrac{\dfrac{\rho}{\vdash A'^{\perp}, A} \qquad \dfrac{\dfrac{\tau}{\vdash A^{\perp}, A} \quad \dfrac{\mu}{\vdash A^{\perp}, A'}\ (cut)}{\vdash A^{\perp}, A'}\ (cut)}{\vdash A'^{\perp}, A'}$$

by idempotence of τ

$$=_{\beta\eta} \dfrac{\tau'}{\vdash A'^{\perp}, A'}$$

by Equation (6) □

Following the proof of Theorem 14, we apply Theorem 2 on τ': we obtain a formula B and proofs π and ϕ respectively of $\vdash A'^{\perp}, B$ and $\vdash B^{\perp}, A'$ such that

$$\dfrac{\tau'}{\vdash A'^{\perp}, A'} \quad =_{\beta\eta} \quad \dfrac{\dfrac{\pi}{\vdash A'^{\perp}, B} \qquad \dfrac{\phi}{\vdash B^{\perp}, A'}}{\vdash A'^{\perp}, A'}\ (cut) \tag{7}$$

Nothing indicates that B is distributed, with good reason: it is easy to build an example where it is not, even though A' is distributed! Nonetheless, the hypotheses on π and ϕ given by Theorem 2 are preserved by isomorphism, using well-chosen representatives of the equivalence classes of the proofs by $=_{\beta\eta}$.

Lemma 25. *Set B' a formula obtained from B by a rewriting among $1 \otimes \cdot \to \cdot$, $\cdot \otimes 1 \to \cdot$, $\perp \parr \cdot \to \cdot$ and $\cdot \parr \perp \to \cdot$ (under context). There exists normal proofs π' and ϕ', respectively of $\vdash A'^{\perp}, B'$ and $\vdash B'^{\perp}, A'$ such that:*

–

$$\dfrac{\tau'}{\vdash A'^{\perp}, A'} \quad =_{\beta\eta} \quad \dfrac{\dfrac{\pi'}{\vdash A'^{\perp}, B'} \qquad \dfrac{\phi'}{\vdash B'^{\perp}, A'}}{\vdash A'^{\perp}, A'}\ (cut) \tag{8}$$

- *no ax-rule in π' (respectively ϕ') uses two sub-formula occurrences of B' (respectively of $B'^{\perp}$);*
- *each 1-rule in π' (respectively ϕ') of main formula from B' (respectively $B'^{\perp}$) is directly above a $\perp$-rule of main formula from $A'^{\perp}$ (respectively A').*

Proof. Assume B' is obtained from B by one application of $1 \otimes D \to D$ – the case of $\perp \parr D \to D$ corresponds to switching π and ϕ in the demonstration, and the two other cases are symmetric. We will show how to obtain π' and ϕ' with their respective hypotheses, and then proceed to the proof of Equation (8).

We first consider how to build π' from π. By hypothesis, in π the 1-rule associated to the rewritten occurrence of 1 from B is directly above a $\perp$-rule

corresponding to a $\bot$ from ${A'}^{\bot}$. Up to rules commutation, the $\otimes$-rule associated to the rewritten occurrence from B appears in π directly below this $\bot$-rule:

$$\dfrac{\dfrac{\dfrac{}{\vdash 1}\,(1)}{\vdash \bot, 1}\,(\bot) \qquad \vdash D, \Gamma}{\vdash 1 \otimes D, \bot, \Gamma}\,(\otimes)$$

Indeed, a $\otimes$-rule on $1 \otimes D$ can be commuted up in the proof until reaching the 1-rule, and we choose to stop just before commuting with the $\bot$-rule. Since rules commutation is included in equality up to cut-elimination [5, Proposition 2.46] and since the hypotheses on π are stable by such rules commutations, we can indeed assume π to be of this shape. We define π' to be π where this part of the proof is replaced by

$$\dfrac{\vdash D, \Gamma}{\vdash D, \bot, \Gamma}\,(\bot)$$

and where in all sequents below we rewrite $1 \otimes D \to D$. Thus, π' is a proof of $\vdash {A'}^{\bot}, B'$, and it is easy to check that the wanted hypotheses on π' hold.

We now proceed similarly to build ϕ' from ϕ. Consider the $\bot$- and $\parr$-rules in ϕ associated to the rewritten occurrences from $B^{\bot}$ in $\bot \parr D^{\bot} \to D^{\bot}$, this being the rewriting in $B^{\bot}$ dual to the rewriting $1 \otimes D \to D$ in B. Up to rules commutation, these two rules appear successively in ϕ:

$$\dfrac{\dfrac{\vdash D^{\bot}, \Delta}{\vdash \bot, D^{\bot}, \Delta}\,(\bot)}{\vdash \bot \parr D^{\bot}, \Delta}\,(\parr)$$

Indeed, a $\bot$-rule is reversible hence can be commuted down in the proof until reaching the introduction of its main formula, allowing to reason as we did for π. We define ϕ' to be ϕ where this part of the proof is replaced by

$$\vdash D^{\bot}, \Delta$$

and where in all sequents below we rewrite $\bot \parr D^{\bot} \to D^{\bot}$. Thus, ϕ' is a proof of $\vdash {B'}^{\bot}, A'$, and it is easy to check that the wanted hypotheses on ϕ' hold.

To conclude the demonstration, it only stays to show Equation (8). To this end, we prove that $\dfrac{\overset{\pi'}{\vdash {A'}^{\bot}, B'} \qquad \overset{\phi'}{\vdash {B'}^{\bot}, A'}}{\vdash {A'}^{\bot}, A'}\,(cut)$ and $\dfrac{\overset{\pi}{\vdash {A'}^{\bot}, B} \qquad \overset{\phi}{\vdash B^{\bot}, A'}}{\vdash {A'}^{\bot}, A'}\,(cut)$ both reduce by cut-elimination to a same proof, and then use Equation (7). To achieve this goal, we build two reduction sequences s and r in parallel, respectively from the first proof and from the second one, such that proofs of s are obtained from proofs of r by the deletion of rules detailed above for π' and ϕ'. This is not overly complex: if a reduction step can be applied on the top-most

cut-rule in the same way on both proofs, we do so. Otherwise, since this strategy is strongly normalising [6, Proposition 3], we must have in the second proof

$$\dfrac{\dfrac{\dfrac{}{\vdash 1}\,(1)}{\vdash \bot, 1}\,(\bot) \qquad \vdash D, \Gamma'}{\vdash 1 \otimes D, \bot, \Gamma'}\,(\otimes) \qquad \dfrac{\dfrac{\vdash D^\bot, \Delta}{\vdash \bot, D^\bot, \Delta'}\,(\bot)}{\vdash \bot \parr D^\bot, \Delta'}\,(\parr)}{\vdash \bot, \Gamma', \Delta'}\,(cut)$$

because this is the only case where the possible cut-elimination steps involve only occurrences absent in the other proof. We reduce it by several steps of cut-elimination into

$$\dfrac{\dfrac{\vdash D, \Gamma'}{\vdash \bot, D, \Gamma'}\,(\bot) \qquad \vdash D^\bot, \Delta}{\vdash \bot, \Gamma', \Delta'}\,(cut)$$

which yields exactly the first proof: we reached a common reduced proof. Thus, Equation (8) holds. □

At present, we apply Lemma 25 repeatedly until reaching a distributed B'. We are now in the setting where we proved Theorem 14 in Section 4: A' and B' are distributed, τ' is an idempotent (by Lemma 24) and we have Equation (8). Therefore, (π', ϕ') is a section-retraction pair. We obtain a section-retraction pair for τ by "passing π' and ϕ' though the isomorphism $A \simeq A'$", the same way τ' was obtained from τ at the start of the present section. Henceforth, we define:

$$\pi'' \quad \stackrel{\text{def}}{=} \quad \dfrac{\overset{\mu}{\vdash A^\bot, A'} \qquad \overset{\pi'}{\vdash A'^\bot, B'}}{\vdash A^\bot, B'}\,(cut) \tag{9}$$

$$\phi'' \quad \stackrel{\text{def}}{=} \quad \dfrac{\overset{\phi'}{\vdash B'^\bot, A'} \qquad \overset{\rho}{\vdash A'^\bot, A}}{\vdash B'^\bot, A}\,(cut) \tag{10}$$

All that is left is showing that this is a section-retraction pair decomposing τ.

Lemma 26. *The pair (π'', ϕ'') is a section-retraction pair.*

Proof. This follows from a simple computation.

$$\begin{aligned} &\dfrac{\overset{\phi''}{\vdash B'^\bot, A} \qquad \overset{\pi''}{\vdash A^\bot, B'}}{\vdash B'^\bot, B'}\,(cut) \\ =_{\beta\eta} &\dfrac{\dfrac{\overset{\phi'}{\vdash B'^\bot, A'} \qquad \overset{\rho}{\vdash A'^\bot, A}}{\vdash B'^\bot, A}\,(cut) \qquad \dfrac{\overset{\mu}{\vdash A^\bot, A'} \qquad \overset{\pi'}{\vdash A'^\bot, B'}}{\vdash A^\bot, B'}\,(cut)}{\vdash B'^\bot, B'}\,(cut) \end{aligned}$$

by Equations (9) and (10)

$$=_{\beta\eta} \dfrac{\phi' \quad \dfrac{\dfrac{\rho \quad \mu}{\vdash A'^{\perp}, A \quad \vdash A^{\perp}, A'}\,(cut) \qquad \pi'}{\vdash A'^{\perp}, A' \qquad \vdash A'^{\perp}, B'}\,(cut)}{\vdash B'^{\perp}, A' \qquad \vdash A'^{\perp}, B'}\,(cut)}{\vdash B'^{\perp}, B'}$$

by several $cut - cut$ commutative cases of cut-elimination

$$=_{\beta\eta} \dfrac{\phi' \quad \dfrac{\dfrac{}{\vdash A'^{\perp}, A'}\,(ax) \qquad \pi'}{\vdash A'^{\perp}, A' \qquad \vdash A'^{\perp}, B'}\,(cut)}{\vdash B'^{\perp}, A' \qquad \vdash A'^{\perp}, B'}\,(cut)}{\vdash B'^{\perp}, B'}$$

by Equation (5)

$$=_{\beta\eta} \dfrac{\phi' \quad \pi'}{\vdash B'^{\perp}, A' \qquad \vdash A'^{\perp}, B'}\,(cut)}{\vdash B'^{\perp}, B'}$$

by an ax key case of cut-elimination

$$=_{\beta\eta} \dfrac{}{\vdash B'^{\perp}, B'}\,(ax)$$

since (π', ϕ') is a section-retraction pair □

Lemma 27. $\vdash A^{\perp}, A \overset{\tau}{} \quad =_{\beta\eta} \quad \dfrac{\vdash A^{\perp}, B' \overset{\pi''}{} \qquad \vdash B'^{\perp}, A \overset{\phi''}{}}{\vdash A^{\perp}, A}\,(cut)$

Proof. This follows from a simple computation.

$$\dfrac{\vdash A^{\perp}, B' \overset{\pi''}{} \qquad \vdash B'^{\perp}, A \overset{\phi''}{}}{\vdash A^{\perp}, A}\,(cut)$$

$$=_{\beta\eta} \dfrac{\dfrac{\vdash A^{\perp}, A' \overset{\mu}{} \qquad \vdash A'^{\perp}, B' \overset{\pi'}{}}{\vdash A^{\perp}, B'}\,(cut) \qquad \dfrac{\vdash B'^{\perp}, A' \overset{\phi'}{} \qquad \vdash A'^{\perp}, A \overset{\rho}{}}{\vdash B'^{\perp}, A}\,(cut)}{\vdash A^{\perp}, A}\,(cut)$$

by Equations (9) and (10)

$$=_{\beta\eta} \dfrac{\vdash A^{\perp}, A' \overset{\mu}{} \qquad \dfrac{\dfrac{\vdash A'^{\perp}, B' \overset{\pi'}{} \qquad \vdash B'^{\perp}, A' \overset{\phi'}{}}{\vdash A^{\perp}, A'}\,(cut) \qquad \vdash A'^{\perp}, A \overset{\rho}{}}{\vdash A^{\perp}, A}\,(cut)}{\vdash A^{\perp}, A}\,(cut)$$

by several $cut - cut$ commutative cases of cut-elimination

$$=_{\beta\eta} \dfrac{\vdash A^{\perp}, A' \overset{\mu}{} \qquad \dfrac{\vdash A'^{\perp}, A' \overset{\tau'}{} \qquad \vdash A'^{\perp}, A \overset{\rho}{}}{\vdash A^{\perp}, A}\,(cut)}{\vdash A^{\perp}, A}\,(cut)$$

by Equation (8)

$$=_{\beta\eta} \dfrac{\dfrac{\mu}{\vdash A^{\perp},A'} \quad \dfrac{\dfrac{\rho}{\vdash A'^{\perp},A} \quad \dfrac{\dfrac{\tau}{\vdash A^{\perp},A} \quad \dfrac{\mu}{\vdash A^{\perp},A'}}{\vdash A^{\perp},A'}\,(cut)}{\vdash A'^{\perp},A'} \quad \dfrac{\rho}{\vdash A'^{\perp},A}}{\vdash A^{\perp},A}\,(cut)}{\vdash A^{\perp},A}\,(cut) \qquad \text{by Equation (6)}$$

$$=_{\beta\eta} \dfrac{\dfrac{\dfrac{\mu}{\vdash A^{\perp},A'} \quad \dfrac{\rho}{\vdash A'^{\perp},A}}{\vdash A^{\perp},A}\,(cut) \quad \dfrac{\tau}{\vdash A^{\perp},A}}{\vdash A^{\perp},A}\,(cut) \quad \dfrac{\dfrac{\mu}{\vdash A^{\perp},A'} \quad \dfrac{\rho}{\vdash A'^{\perp},A}}{\vdash A^{\perp},A}\,(cut)}{\vdash A^{\perp},A}\,(cut)$$

by several $cut - cut$ commutative cases of cut-elimination

$$=_{\beta\eta} \dfrac{\dfrac{}{\vdash A^{\perp},A}\,(ax) \quad \dfrac{\tau}{\vdash A^{\perp},A}}{\vdash A^{\perp},A}\,(cut) \quad \dfrac{}{\vdash A^{\perp},A}\,(ax)}{\vdash A^{\perp},A}\,(cut) \qquad \text{by Equation (4)}$$

$$=_{\beta\eta} \dfrac{\tau}{\vdash A^{\perp},A} \qquad \text{by two } ax \text{ key cases of cut-elimination} \quad \square$$

Therefore, Lemmas 26 and 27 together imply that τ is a split idempotent. This handles the claims in the proof of Theorem 14 in Section 4 that we can assume the formulas A and B to be distributed.